

# Multiwavelength Analysis of X-ray Binary Outbursts

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A Project Report By

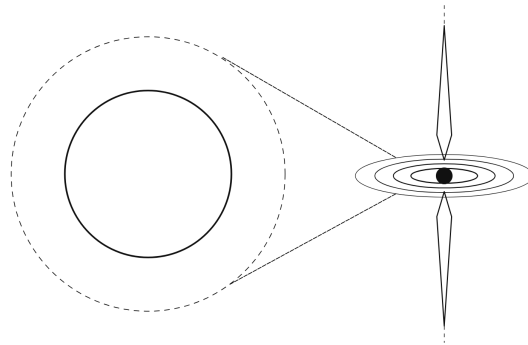
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# 1 Introduction

X-ray binaries (XRBs) are gravitationally bound two-body systems comprising a compact primary — either a neutron star (NS) or a stellar-mass black hole (BH) — and a non-degenerate companion (donor) star. As matter is transferred from the donor toward the compact object, gravitational potential energy is converted into thermal and non-thermal radiation predominantly in the X-ray band, rendering these systems among the most luminous persistent and transient X-ray sources in the Galaxy [18, 31]. Their study spans high-energy astrophysics, general relativity, and nuclear physics, making XRBs uniquely suited for probing physical regimes inaccessible in terrestrial laboratories.

## 1.1 The Compact Primary

### Neutron Stars

Neutron stars are the remnants of core-collapse supernovae, with masses in the range  $1.4\text{--}2.1 M_\odot$  and radii of approximately 10–12 km. The presence of a solid surface, strong magnetic fields ( $B \sim 10^8\text{--}10^{12}$  G), and the capacity for thermonuclear ignition of accreted material give rise to observational signatures — most notably Type-I X-ray bursts — that are absent in BH systems (Section 1.5).

### Stellar-Mass Black Holes

Stellar-mass black holes span masses of approximately  $5\text{--}20 M_\odot$  [26]. Unlike NSs, they possess no physical surface; the innermost stable circular orbit (ISCO), whose radius is set by the dimensionless spin parameter  $a^* \in [0, 1]$ , defines the effective inner boundary of the accretion disk. Spectral modelling of the thermal disk emission and relativistic reflection features provide observational constraints on  $a^*$  [27].

## 1.2 Energetics of Accretion

The dominant energy release mechanism in XRBs is the conversion of gravitational potential energy. For a test mass  $m$  accreted onto a compact object of mass  $M$  and radius  $R$ , the available energy is

$$E = \frac{GMm}{R}. \quad (1)$$

Expressed as an efficiency  $\eta \equiv E/mc^2$ , accretion onto a NS yields  $\eta \approx 10\%$ , while accretion onto a maximally spinning BH can reach  $\eta \approx 42\%$  [8]. Both values substantially exceed the efficiency of hydrogen fusion ( $\eta \approx 0.7\%$ ), which accounts for the extreme luminosities ( $L_X \sim 10^{37}\text{--}10^{38} \text{ erg s}^{-1}$ ) sustained at relatively low mass-transfer rates ( $\dot{M} \sim 10^{-10}\text{--}10^{-8} M_\odot \text{ yr}^{-1}$ ).

## 1.3 Mass Transfer and Accretion Disk Physics

### Roche-Lobe Overflow

When the donor star evolves to fill its Roche lobe, material flows through the inner Lagrangian point  $L_1$  under the combined gravitational and centrifugal potential [23]. The specific angular

momentum of the transferred gas is too large for direct radial infall; it circularises into a Keplerian accretion disk.

### Stellar Wind Accretion

In systems with massive OB donors, the compact object intercepts a fraction of the supersonic stellar wind within the Bondi–Hoyle–Lyttleton accretion radius [22]. This process does not necessarily produce a persistent disk, and the resulting accretion rate is generally lower and more variable than in Roche-lobe overflow systems.

### Disk Structure and Viscosity

In Roche-lobe overflow systems, the disk is optically thick and geometrically thin (the standard  $\alpha$ -disk; Shakura and Sunyaev 29). Angular momentum is transported outward by turbulent viscosity, parameterised by the dimensionless Shakura–Sunyaev  $\alpha$  parameter, allowing steady mass inflow. The effective temperature profile of an unirradiated disk follows

$$T(r) \propto r^{-3/4}, \quad (2)$$

so the innermost disk regions ( $r \sim 10^6$ – $10^7$  cm) reach X-ray temperatures, while the outer disk radiates in the UV and optical bands.

## 1.4 Classification of X-Ray Binaries

XRBs are conventionally divided according to the mass of the donor star, which governs the dominant mass-transfer mechanism, the evolutionary age of the system, and its spatial distribution in the Galaxy [10, 19].

### Low-Mass X-Ray Binaries

In low-mass X-ray binaries (LMXBs), the donor mass satisfies  $M_{\text{donor}} \lesssim 1 M_{\odot}$ ; donors are typically late-type main-sequence stars (K/M dwarfs), subgiants, or white dwarfs. Mass transfer proceeds exclusively via Roche-lobe overflow [32]. Optical emission is dominated by reprocessed X-rays in the disk, resulting in faint counterparts ( $V \gtrsim 15$  mag) that are difficult to study independently of the X-ray flux. LMXBs are old systems ( $\sim 10^9$ – $10^{10}$  yr) concentrated in the Galactic Bulge and globular clusters. The majority are transient, spending extended intervals in quiescence ( $L_X \sim 10^{30}$ – $10^{33}$  erg s $^{-1}$ ) interrupted by bright accretion outbursts.

### High-Mass X-Ray Binaries

High-mass X-ray binaries (HMXBs) host donors with  $M_{\text{donor}} \gtrsim 10 M_{\odot}$ . Accretion proceeds either through dense stellar winds from OB supergiants or via a circumstellar decretion disk in Be/X-ray binary systems (BeXBs), where outbursts are triggered at orbital periastron [25]. The donor luminosity dominates the optical flux ( $V \lesssim 12$  mag). HMXBs are young systems ( $\sim 10^6$ – $10^7$  yr) concentrated along the Galactic Plane. Notable subclasses include supergiant

X-ray binaries (sg-HMXBs) and supergiant fast X-ray transients (SFXTs), which display brief, intense flares whose origin remains debated [21].

### Other Populations

Intermediate-mass X-ray binaries (IMXBs;  $1 \lesssim M_{\text{donor}}/M_{\odot} \lesssim 10$ ) are rare due to the short timescale of thermal-timescale mass transfer [24]. Ultraluminous X-ray sources (ULXs) are non-nuclear point sources with isotropic luminosities exceeding the Eddington limit for a  $10 M_{\odot}$  black hole ( $L_X > 10^{39} \text{ erg s}^{-1}$ ), and are attributed to either super-Eddington accretion or, in some cases, accretion onto intermediate-mass black holes [13].

## 1.5 Observational Signatures: Bursts and Outbursts

Two classes of high-energy transient events are observed in XRBs. Their physical origins are distinct, and the distinction is diagnostically important for identifying the compact object type.

### Type-I X-Ray Bursts

Type-I X-ray bursts arise from unstable thermonuclear burning of accreted hydrogen and helium on the solid surface of a NS. As material accumulates, the base of the accreted layer reaches conditions sufficient to ignite the triple- $\alpha$  process and the rapid-proton capture (*rp*) process, producing a thermonuclear runaway [18, 30]. The resulting burst exhibits a rapid flux rise ( $\sim 1\text{--}10$  s), an exponential decay ( $\sim 10\text{--}100$  s) as the burning layer cools, and recurrence times of hours to days depending on the accretion rate. Because a solid surface is an absolute prerequisite, Type-I bursts provide unambiguous identification of a NS primary; no equivalent phenomenon is possible in BH systems.

### X-Ray Outbursts

X-ray outbursts are episodes of dramatically enhanced accretion lasting weeks to months, separated by quiescent intervals of years to decades. They manifest as correlated brightening across radio, optical, UV, and X-ray wavelengths and are observed in both NS and BH binary systems [2]. The underlying mechanism is a thermal-viscous instability in the accretion disk, described in the following subsection.

## 1.6 Outburst Mechanism and Spectral State Evolution

### The Disk Instability Model

The disk instability model (DIM) provides the standard theoretical framework for transient XRB outbursts [11, 17]. Opacity in the disk changes sharply near the hydrogen ionisation temperature ( $T \sim 6\,000\text{--}10\,000$  K), leading to a thermal-viscous instability. The system undergoes a limit cycle between two stable branches in the surface density–temperature ( $\Sigma\text{--}T$ ) plane: a cold, neutral branch with low viscosity ( $\alpha_{\text{cold}}$ ) corresponding to quiescence, and a hot, ionised branch

with higher viscosity ( $\alpha_{\text{hot}}$ ) corresponding to outburst. Irradiation of the outer disk by X-rays from the inner accretion flow stabilises the disk at intermediate radii, prolonging outburst duration relative to unirradiated DIM predictions [5].

### Persistent and Transient Behaviour

Systems in which the long-term mean accretion rate exceeds the critical DIM threshold maintain a continuously ionised disk and are observed as persistent sources ( $L_X \sim 10^{37}\text{--}10^{38} \text{ erg s}^{-1}$ ). Transient systems accrete below this threshold on average; the disk recombines and cools in quiescence ( $L_X \sim 10^{30}\text{--}10^{33} \text{ erg s}^{-1}$ ) until an instability front propagates inward, triggering an outburst that can approach the Eddington luminosity  $L_{\text{Edd}}$  [14].

### Spectral States

The X-ray spectral and timing properties of a BH binary evolve systematically through an outburst and are classified into canonical spectral states [1, 26]:

- **Hard state.** The spectrum is dominated by a hard power law produced by inverse Compton scattering in a hot, optically thin corona. A compact, flat-spectrum relativistic radio jet is present.
- **Soft state.** The spectrum is dominated by quasi-thermal emission from an optically thick, geometrically thin accretion disk. The radio jet is suppressed; thermally driven disk winds may be present.

State transitions follow a well-defined path in the hardness–intensity diagram (HID), tracing a counter-clockwise “q”-shaped hysteresis loop over a complete outburst cycle [7]. Notably, the hard-to-soft transition consistently occurs at higher luminosity than the reverse transition; this hysteresis is not fully reproduced by current theoretical models and remains an open problem [6].

## 2 Instruments and Databases

This project draws on data from multiple space-based and ground-based observatories, as well as several astronomical catalogues and broker platforms. The instruments provided the raw observational data — primarily in X-ray and optical bands — while the databases and catalogues were used to cross-identify sources, retrieve archival light curves, and classify candidates. The X-ray instruments cover the soft to hard X-ray regime, critical for tracking spectral state transitions in black hole X-ray binaries, while the optical survey provides contemporaneous coverage of outburst-driven changes in the accretion disc and jet emission. Below is a summary of all resources used in this project.

### Telescopes and Instruments:

- *Neil Gehrels Swift Observatory* – Burst Alert Telescope (BAT)
- *Monitor of All-sky X-ray Image* (MAXI)
- *Zwicky Transient Facility* (ZTF)

### Databases and Catalogues:

- SIMBAD Astronomical Database
- BlackCAT – Black Hole X-ray Binary Catalogue [3]
- VizieR Catalogue Service [19]
- Alerce ZTF Broker

## 2.1 Telescopes and Instruments

### 2.1.1 Neil Gehrels Swift Observatory – Burst Alert Telescope (BAT)

The *Neil Gehrels Swift Observatory*, launched in November 2004, is a NASA multi-wavelength space telescope designed primarily for the rapid detection and follow-up of gamma-ray bursts [9]. The Burst Alert Telescope (BAT) is the wide-field, hard X-ray instrument onboard Swift and is the primary tool used in this project for monitoring X-ray binary activity over long timescales.

BAT operates in the energy range of **14–195 keV**, placing it firmly in the hard X-ray regime where emission from accreting compact objects — Comptonised radiation from the corona and power-law continuum from the jet — dominates. It carries a large detector plane of  $\sim 5200 \text{ cm}^2$  composed of 32,768 CdZnTe (cadmium zinc telluride) detector elements, each sensitive to individual photon events.

**Coded Aperture Mask and Imaging Principle.** BAT uses a *coded aperture mask* (CAM) to achieve imaging in the hard X-ray band, where conventional focusing optics are not feasible. The mask consists of a partially opaque grid of lead tiles arranged in a pseudo-random pattern, placed  $\sim 1 \text{ m}$  above the detector plane. Incoming photons from a source at a given sky position cast a unique shadow pattern — determined by the mask geometry — onto the detector array.

Because each sky direction projects a distinct shadow, the sky image is reconstructed by a cross-correlation (deconvolution) of the measured detector count map with the known mask pattern. This technique, known as coded-mask imaging, allows BAT to simultaneously image a wide field of view of **1.4 sr** (partially coded) without mechanically slewing the telescope, making it ideal for untargeted all-sky monitoring.

The angular resolution achieved through this method is  $\sim$ **1-4 arcmin** (90% containment radius), which is sufficient for unambiguous source identification of bright X-ray binaries and transients, though not for resolving crowded fields. The energy resolution is  $\sim$ **7 keV FWHM** at 122 keV, adequate for broad spectral classification (hard vs. soft state identification) but not for fine spectral line analysis. The coded mask also provides partial shielding against cosmic ray background, improving sensitivity for faint sources in survey mode.

The BAT hard X-ray transient monitor provides publicly available, near-real-time light curves for  $\sim$ 1000 known hard X-ray sources, updated daily, and is archived at *HEASARC* (High Energy Astrophysics Science Archive Research Center) hosted at NASA’s Goddard Space Flight Center [16]. These light curves form the backbone of long-term hard X-ray monitoring used in this project.

### 2.1.2 Monitor of All-sky X-ray Image (MAXI)

MAXI is a Japanese X-ray all-sky monitor installed on the *Japanese Experiment Module – Exposed Facility* (JEM-EF) of the International Space Station (ISS) [20]. Operational since August 2009, it is maintained by RIKEN (Institute of Physical and Chemical Research), Japan, in collaboration with JAXA (Japan Aerospace Exploration Agency). MAXI data are publicly hosted at <http://maxi.riken.jp>.

MAXI carries two slit-camera detector systems. The *Gas Slit Camera* (GSC) is the primary X-ray monitor, using position-sensitive proportional counters sensitive across the following energy bands:

- 2–4 keV (soft X-ray)
- 4–10 keV (medium X-ray)
- 10–20 keV (hard X-ray)
- 2–20 keV (full band)

The *Solid State Slit Camera* (SSC) operates at softer energies (0.7–7 keV) using CCD detectors, extending MAXI’s bandpass toward the extreme soft X-ray regime. The GSC scans the entire sky every  $\sim$ 92 minutes — one ISS orbital period — providing near-continuous all-sky monitoring with no pointed observing constraints. Its angular resolution is  $\sim$ **1.5°** (FWHM) and energy resolution  $\sim$ **18% at 6 keV** (FWHM), characteristic of gas proportional counter technology.

**Role in Outburst Detection.** The continuous all-sky coverage of MAXI makes it uniquely suited for detecting and tracking X-ray binary outbursts in real time. When a quiescent black hole X-ray binary (BHXB) enters an outburst — driven by a thermal-viscous instability in the

accretion disc — its X-ray flux can rise by several orders of magnitude over days to weeks. MAXI/GSC detects this flux increase across multiple energy bands simultaneously, allowing early characterisation of spectral hardness and the identification of the initial hard state, which is typically the first phase of an outburst. MAXI issues public *X-ray transient alert notices* (via the MAXI mailing list and the Astronomer’s Telegram) within hours of detecting a significant flux increase, enabling rapid multi-wavelength follow-up by the broader community.

Beyond single-outburst detection, MAXI provides long-term light curves spanning over a decade, which are invaluable for identifying recurrent transients, characterising outburst recurrence intervals, and measuring peak luminosities across different outburst episodes. The MAXI on-demand data analysis system allows users to extract custom-binned light curves and spectra for any sky position, making it a highly flexible monitoring resource. In this project, MAXI light curves were used to track outburst onset, evolution, and state transitions for candidate BHXBs identified in the sample.

### 2.1.3 Zwicky Transient Facility (ZTF)

The *Zwicky Transient Facility* (ZTF) is a ground-based optical time-domain survey operating at *Palomar Observatory*, California, managed by Caltech in collaboration with multiple international partners. ZTF became fully operational in 2018 and is funded in part by the U.S. National Science Foundation (NSF) and an international partnership.

ZTF uses a wide-field mosaic camera mounted on the 48-inch Samuel Oschin Telescope, covering a field of view of **47 deg<sup>2</sup>** per exposure. It surveys the sky in three optical filters: *g*-band ( $\sim 472$  nm), *r*-band ( $\sim 633$  nm), and *i*-band ( $\sim 789$  nm), covering  $\sim \mathbf{15,000\ deg^2}$  of the northern sky per night. The survey cadence is  $\sim \mathbf{2\ nights}$  between revisits in a single filter under standard survey operations, with the limiting magnitude reaching  $\sim \mathbf{20.8\ mag}$  in the *r*-band under median conditions. The plate scale is **1.01 arcsec/pixel**, with effective angular resolution set by typical Palomar seeing ( $\sim 2$  arcsec FWHM). ZTF is a photometric survey and does not provide spectral resolution directly; spectroscopic follow-up requires separate facilities.

**Alert Pipeline Architecture.** Every ZTF exposure is processed in near-real-time by the *ZTF Science Data System* (ZSDS) at the Infrared Processing and Analysis Center (IPAC), Caltech. The pipeline performs bias subtraction, flat-fielding, and astrometric calibration before running image subtraction against deep co-added reference images built from prior ZTF data. Sources detected in the subtracted image at  $\geq 5\sigma$  significance are extracted as *difference-image detections* and packaged into structured *alert packets* in Apache Avro format. Each alert packet contains: the source position and photometry from the current epoch, the preceding 30-day light curve history, a  $63 \times 63$  arcsec cutout of the science, reference, and difference images, and a set of real-bogus quality scores derived from a machine-learning classifier trained to separate genuine astrophysical transients from artefacts.

These alert packets are distributed publicly in real time via the *ZTF alert distribution system*, which broadcasts  $\sim 10^6$  alerts per night over a Kafka message queue. This stream is ingested by community broker platforms — including Alerce (Section 2.2) — which apply further filtering, cross-matching, and classification. The full ZTF alert archive is also accessible through the *IRSA* (NASA/IPAC Infrared Science Archive). In this project, ZTF optical light curves were

retrieved via Alerce for cross-matching with X-ray outburst epochs identified through MAXI and BAT.

## 2.2 Databases and Catalogues

### 2.2.1 SIMBAD Astronomical Database

SIMBAD (*Set of Identifications, Measurements, and Bibliography for Astronomical Data*) is a reference astronomical database maintained and operated by the *Centre de Données astronomiques de Strasbourg* (CDS), France . It provides cross-identification of astronomical objects across catalogues and publications, along with basic data including coordinates, photometry, spectral types, proper motions, radial velocities, and bibliographic references compiled from the peer-reviewed literature.

In this project, SIMBAD was used to confirm source identities, retrieve precise coordinates, and cross-match candidate X-ray binaries with known catalogued objects across multiple wavelength regimes. The database is freely accessible at <https://simbad.u-strasbg.fr> and supports manual queries as well as programmatic access via the TAP/ADQL interface and the `astroquery.simbad` Python module.

### 2.2.2 BlackCAT – Black Hole X-ray Binary Catalogue

BlackCAT is a web-based, living catalogue of confirmed and candidate stellar-mass black hole X-ray binaries (BHXBs), compiled and maintained by Corral-Santana et al. [4]. It represents one of the most complete and regularly updated references for the Galactic BHXB population, drawing exclusively from peer-reviewed literature to ensure reliability.

**Catalogue Parameters.** For each source, BlackCAT provides a comprehensive set of measured and derived parameters including:

- Source name and known aliases across catalogues
- Celestial coordinates (RA, Dec) and Galactic coordinates ( $l$ ,  $b$ )
- Orbital period  $P_{\text{orb}}$  and, where available, the mass function  $f(M)$ , which sets a lower limit on the compact object mass
- Black hole mass estimate  $M_{\text{BH}}$  and donor (companion) star mass  $M_2$ , derived from dynamical studies
- Orbital inclination  $i$ , which affects the observed flux and spectral appearance
- Distance estimate  $d$  and the associated hydrogen column density  $N_{\text{H}}$ , a proxy for line-of-sight absorption
- Donor star spectral type, which constrains the nature of mass transfer (Roche lobe overflow vs. stellar wind)
- Outburst history: dates of known outbursts, peak X-ray flux, and outburst classification where available

- Transient vs. persistent classification

This parameter set makes BlackCAT not merely an identification list but a quantitative reference for population studies. In this project, BlackCAT was used as the primary reference for identifying known BHXBs within the sample, verifying source classifications, and retrieving orbital and distance parameters needed for luminosity calculations.

### 2.2.3 VizieR Catalogue Service

VizieR is a large-scale astronomical catalogue access service maintained by the CDS in Strasbourg, France, in partnership with several international data centres including the NASA Astrophysics Data System (ADS) mirror nodes and the Japanese and Canadian CDS affiliates. It is one of the most comprehensive repositories of published astronomical data, hosting over 22,000 catalogues and associated data tables extracted directly from peer-reviewed journal articles. Each catalogue in VizieR is linked to its source publication, ensuring full traceability between data and literature.

VizieR supports a range of query modes including positional cone searches, parameter range filtering, and full-catalogue downloads, accessible via a web interface at <https://vizier.cds.unistra.fr> or programmatically through its TAP (Table Access Protocol) endpoint and the `astroquery.vizier` Python module. Catalogue data are stored in standardised tabular formats and served in multiple output formats (VOTable, FITS, CSV, ASCII). In this project, VizieR was used to retrieve source catalogues relevant to X-ray binary identification and classification, in particular the catalogue of Liu et al. 19, which was accessed directly through the VizieR interface for cross-matching purposes.

### 2.2.4 ALeRCE ZTF Broker

Alerce is an open, community-driven astronomical broker platform developed and maintained by the *Millennium Institute of Astrophysics* (MAS), Chile, with support from the Chilean Ministry of Economy. It is purpose-built to ingest, process, and serve the real-time public alert stream produced by ZTF, and will serve as one of the primary community brokers for the forthcoming Legacy Survey of Space and Time (LSST).

**Alert Stream Architecture.** Alerce continuously ingests the ZTF Kafka alert stream — approximately  $10^6$  alerts per night — in real time. Upon ingestion, each Avro-format alert packet is unpacked and passed through a multi-stage processing pipeline. In the first stage, alerts are cross-matched against major astronomical catalogues (including SIMBAD, Gaia, AllWISE, and the PS1 source catalogue) to attach known-source context to each detection. In the second stage, a suite of machine-learning classifiers is applied, including:

- A *real-bogus classifier* to reject image subtraction artefacts
- A *stamp classifier* operating on the difference-image cutouts using a convolutional neural network (CNN) to separate point sources, extended sources, and bogus detections

- A *light curve classifier* using recurrent neural networks (RNNs) trained on multi-band ZTF photometry to assign probabilistic class labels (e.g., supernova, active galactic nucleus, variable star, transient)

Processed alerts, along with their cross-match results and classifier outputs, are stored in the Alerce database and made available through a web interface at <https://alerce.online> and a public REST API. Users can query by sky position, filter by classifier probability, and retrieve the full multi-epoch photometric light curve for any ZTF-detected source. The platform retains the complete ZTF alert history, making it an effective archive for retrospective time-domain studies. In this project, Alerce was used to retrieve ZTF optical light curves for X-ray binary candidates identified through MAXI and BAT monitoring, enabling multi-wavelength characterisation of outburst events.

### 3 Cross Matching Across Databases

This section describes the multi-step cross-matching pipeline developed to identify X-ray Binary (XRB) systems with simultaneous coverage in four independent astronomical databases:

- **BAT** — the *Swift*/BAT 157-month hard X-ray catalog (input source list, stored as a FITS file).
- **MAXI** — the Monitor of All-sky X-ray Image (RIKEN) archive, queried via HTTP to confirm a source has a known lightcurve page.
- **SIMBAD** — the CDS astronomical database, used to obtain precise optical coordinates and object classifications.
- **ZTF** — the Zwicky Transient Facility, accessed via the ALeRCE broker API, for optical transient/variability data.
- **VizieR** — the CDS catalogue service, queried via `astroquery` to access the Liu et al. [19] LMXB and HMXB reference catalogs
- **BlackCAT** [3] — a curated online catalog of black hole X-ray transients (Corral-Santana et al. 2016), cross-matched by source name and sky position to identify black hole systems.

And In the later section we will see the use of and

The ultimate goal is to build a clean sample of XRBs that have confirmed MAXI outburst detections *and* sufficient ZTF optical photometry within the same time window, enabling multi-wavelength outburst studies.

#### 3.1 Cross-matching Swift/BAT Sources with MAXI, SIMBAD, and ZTF to Identify X-ray Binaries with Optical Counterparts

The goal of this pipeline is to identify confirmed X-ray binaries (XRBs) that are simultaneously detected by the Swift/BAT hard X-ray survey, monitored by MAXI in the soft-to-medium X-ray band, and have an optical counterpart archived in the Zwicky Transient Facility (ZTF). The pipeline proceeds in four sequential cross-matching steps: BAT to MAXI, MAXI to SIMBAD, SIMBAD to ZTF, and a final XRB-type filter. The output is a master catalog of XRBs with multi-band coverage suitable for outburst analysis.

##### Databases

**Swift/BAT.** The Neil Gehrels Swift Observatory’s Burst Alert Telescope (BAT) hard X-ray survey catalog (`BAT_catalog.fits`) provides source positions, names, and alternate identifiers for hard X-ray emitters detected in the 14–195 keV band.

**MAXI.** The Monitor of All-sky X-ray Image (MAXI) Gas Slit Camera (GSC) provides daily-cadence soft X-ray light curves (2–20 keV) served as plain-text `.dat` files via a public web interface. Source pages are identified by a positionally-derived J-name string.

**SIMBAD.** The SIMBAD astronomical database at CDS Strasbourg supplies standardised object-type classifications (`otype`) and precise optical/X-ray coordinates for each source name query.

**ZTF via ALerCE.** The Zwicky Transient Facility (ZTF) optical survey is queried through the ALerCE broker API, which provides cone-search access to ZTF object identifiers, mean coordinates, and multi-band light curves (g, r, i bands).

### Step 1: BAT to MAXI Cross-match

Each BAT source position (R.A., Dec. in degrees) is converted to a MAXI J-name string by the following coordinate transformation. The right ascension is first converted to hours and split into integer hours  $H$  and minutes  $M$  (truncated, not rounded). The declination sign and absolute value yield the degrees  $D$  and tenths  $d$ . The resulting identifier takes the form:

$$J + HHMM + \text{sign} + DDd \quad (3)$$

where all components are zero-padded to their respective widths.

The MAXI web server is queried with this identifier to check whether a source page exists. An HTTP 200 response confirms a MAXI detection. A fixed delay of 0.3 s between requests is applied to avoid overloading the server. Sources returning a 200 response are flagged as MAXI-matched and their BAT names, positions, and derived MAXI identifiers are saved to `BAT_MAXI_matched_sources.csv`.

### Step 2: MAXI-matched Sources to SIMBAD

Each BAT catalogue name from the MAXI-matched list is submitted to SIMBAD as a named-object query. The query returns the SIMBAD main identifier, object type (`otype`), and precise coordinates (R.A., Dec. in degrees). The object type is used in the final XRB filter (Section below). The SIMBAD coordinates replace the BAT positions in all downstream steps because they are derived from dedicated optical or X-ray astrometry and are typically more precise than the BAT localisation. Results are written to a CSV file.

A representative query loop is shown below:

```
custom_simbad = Simbad()
custom_simbad.add_votable_fields('otype')

for index, row in matched_df.iterrows():
    result_table = custom_simbad.query_object(row['BAT_Catalogue_Name'])
    if result_table is not None:
        simbad_main_ids.append(result_table['main_id'][0])
        simbad_otypes.append(result_table['otype'][0])
        simbad_ras.append(float(result_table['ra'][0]))
        simbad_decs.append(float(result_table['dec'][0]))
```

Sources for which SIMBAD returns no result are retained in the table with null coordinate entries and are excluded from the ZTF cross-match step via a `dropna` filter on the coordinate columns.

### Step 3: SIMBAD Positions to ZTF via ALeRCE

The SIMBAD-resolved coordinates are used to query the ZTF database through the ALeRCE broker API with a cone-search radius of 5 arcsec. This radius is chosen to match the typical positional uncertainty of optical counterparts to XRBs. If multiple ZTF objects fall within the cone, the one with the smallest angular separation from the SIMBAD position is selected:

```
matches = client.query_objects(survey='ztf',
                               ra=ra_simbad, dec=dec_simbad,
                               radius=5.0)          # arcseconds

center = SkyCoord(ra=ra_simbad, dec=dec_simbad, unit='deg')
ztf_coords = SkyCoord(ra=matches['meanra'].values,
                      dec=matches['meandec'].values, unit='deg')
seps = center.separation(ztf_coords).arcsec
best = matches.iloc[np.argmin(seps)]
```

The ZTF object identifier, mean coordinates, and angular separation from the SIMBAD position are appended to the table. The result is saved to `BAT_MAXI_SIMBAD_ZTF_match.csv`.

### Step 4: XRB-type Filter

The full matched table is filtered to retain only sources whose SIMBAD object type is LXB (low-mass X-ray binary) or HXB (high-mass X-ray binary), and for which their optical (ZTF) counterpart was successfully identified. All other source types (e.g. active galactic nuclei, cataclysmic variables, coronal emitters) are discarded.

```
xrb_df = df[
    (df['SIMBAD_0type'].isin(['LXB', 'HXB'])) &
    (df['ZTF_Object_ID'].notna())
].copy()
```

The surviving sources are written to a csv file, whose columns are summarised in Table 1. This file constitutes the master XRB catalog used in all subsequent stages of the analysis.

### Cross-match Summary

Table 2 summarises the source counts at each stage of the pipeline. Numbers in the last column are to be filled with the values printed by the pipeline at runtime.

The final XRB catalog from Step 4 is used as input to the outburst detection stage, where MAXI daily light curves are analysed for flux excursions above a sigma-clipped baseline threshold.

The list of these 20 **Gold** sources is given in the Table 3.

Table 1: Column schema of the final cross-matched XRB catalog.

| Column             | Description                               |
|--------------------|-------------------------------------------|
| BAT_Catalogue_Name | Source name from the Swift/BAT catalog    |
| RA_BAT             | Right ascension from BAT (degrees, J2000) |
| DEC_BAT            | Declination from BAT (degrees, J2000)     |
| MAXI_Name          | MAXI J-name derived from BAT position     |
| SIMBAD_ID          | SIMBAD main identifier                    |
| SIMBAD_Otype       | SIMBAD object type (LXB or HXB)           |
| SIMBAD_RA          | Precise R.A. from SIMBAD (degrees, J2000) |
| SIMBAD_DEC         | Precise Dec. from SIMBAD (degrees, J2000) |
| ZTF_Object_ID      | ZTF object identifier from ALeRCE         |

Table 2: Source counts at each cross-matching stage.

| Step | Stage                                      | Sources |
|------|--------------------------------------------|---------|
| 0    | BAT input catalog                          | 932     |
| 1    | BAT $\cap$ MAXI (HTTP 200 match)           | 270     |
| 2    | + SIMBAD resolution (coordinates returned) | 266     |
| 3    | + ZTF counterpart within 5 arcsec          | 86      |
| 4    | + XRB filter (LXB or HXB only)             | 20      |

Followed by Light Curves 1 of selected sources which has substantial optical data from ZTF.

Table 3: The cross matched XRB list between BAT, MAXI, SIMBAD and ZTF, The "Gold" sources

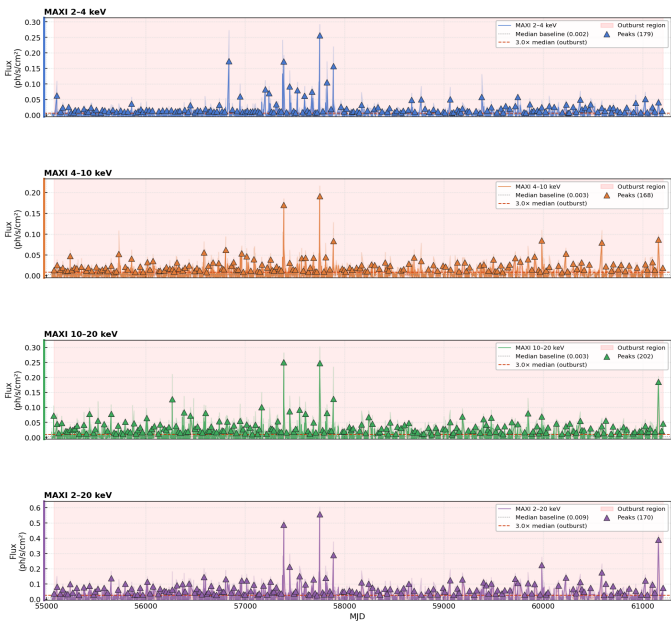
| S.No. | BAT ID             | RA(BAT)  | DEC(BAT) | MAXI Name | SIMBAD ID             | XRB Type | RA(SIMBAD) | DEC(SIMBAD) | ZTF ID       |
|-------|--------------------|----------|----------|-----------|-----------------------|----------|------------|-------------|--------------|
| 1     | 4U 0115+634        | 19.63    | 63.74    | J0118+637 | V* V635 Cas           | HXB      | 19.6332    | 63.7425     | ZTF18aczqqgn |
| 2     | IGR J06074+2205    | 91.85    | 22.083   | J0607+220 | IGR J06074+2205       | HXB      | 91.8609    | 22.0966     | ZTF17aaarcit |
| 3     | 4U 0614+09         | 94.28    | 9.137    | J0617+091 | V* V1055 Ori          | LXB      | 94.2807    | 9.1371      | ZTF18acukluh |
| 4     | 1A 0620-00         | 95.675   | -0.35    | J0622-003 | 1A 0620-00            | HXB      | 95.6856    | -0.3456     | ZTF18acszzdo |
| 5     | MXB 0656-072       | 104.612  | -7.263   | J0658-072 | 2E 1752               | HXB      | 104.572    | -7.2098     | ZTF18acpdmgx |
| 6     | Sco X-1            | 244.979  | -15.64   | J1619-156 | V* V818 Sco           | LXB      | 244.9795   | -15.6403    | ZTF19aetdak  |
| 7     | Her X-1            | 254.457  | 35.342   | J1657+353 | V* HZ Her             | LXB      | 254.4575   | 35.3424     | ZTF18aabpzc  |
| 8     | SAX J1819.3-2525   | 274.825  | -25.417  | J1819-254 | V* V4641 Sgr          | HXB      | 274.8401   | -25.4072    | ZTF21abvdujh |
| 9     | SAX J1828.5-1037   | 277.138  | -10.63   | J1828-106 | 2XMM J182833.7-103700 | LXB      | 277.1407   | -10.6177    | ZTF18abhjypr |
| 10    | IGR J18410-0535    | 280.252  | -5.596   | J1841-055 | IGR J18410-0535       | HXB      | 280.2518   | -5.5962     | ZTF19aonvju  |
| 11    | XTE J1855-026      | 283.8804 | -2.6067  | J1855-026 | XTE J1855-026         | HXB      | 283.8767   | -2.6046     | ZTF24aesxju  |
| 12    | XTE J1859+226      | 284.673  | 22.658   | J1858+226 | V* V406 Vul           | HXB      | 284.6729   | 22.6583     | ZTF21aagyvqr |
| 13    | H 1907+097         | 287.4    | 9.833    | J1909+098 | 4U 1907+09            | HXB      | 287.4085   | 9.8298      | ZTF25aaeumpl |
| 14    | Aql X-1            | 287.825  | 0.583    | J1911+005 | X Aql X-1             | LXB      | 287.8169   | 0.585       | ZTF18acceda  |
| 15    | SS 433             | 287.956  | 4.99     | J1911+049 | SS 433                | HXB      | 287.9565   | 4.9827      | ZTF18abcgtnb |
| 16    | SWIFT J1922.7-1716 | 290.679  | -17.2827 | J1922-172 | 2MAXI J1922-171       | LXB      | 290.6542   | -17.2842    | ZTF22abfbyjc |
| 17    | KS 1947+300        | 297.39   | 30.213   | J1949+302 | KS 1947+300           | HXB      | 297.3979   | 30.2088     | ZTF18accozaw |
| 18    | 4U 1957+115        | 299.851  | 11.709   | J1959+117 | V* V1408 Aql          | LXB      | 299.85     | 11.7083     | ZTF21abeihmb |
| 19    | GRO J2058+42       | 314.698  | 41.7769  | J2058+417 | GRO J2058+42          | HXB      | 314.6981   | 41.777      | ZTF18abcbth  |
| 20    | Cyg X-2            | 326.172  | 38.322   | J2144+383 | V* V1341 Cyg          | LXB      | 326.1715   | 38.3214     | ZTF19aavqbvs |

# Multi-band X-ray & Optical Lightcurves — Selected XRB Sources

## BAT · MAXI (2-4, 4-10, 10-20, 2-20 keV) · ZTF (g, r, i)

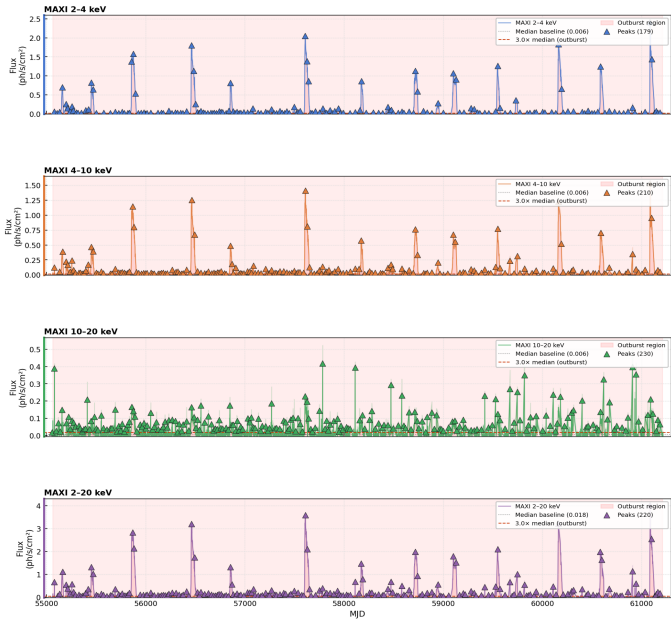
### IGR J06074+2205

IGR J06074+2205  
MAXI ID: J06074-2205 · ZTF ID: ZTF18aacrit



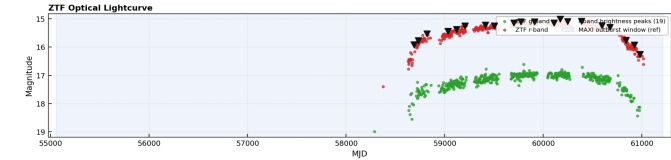
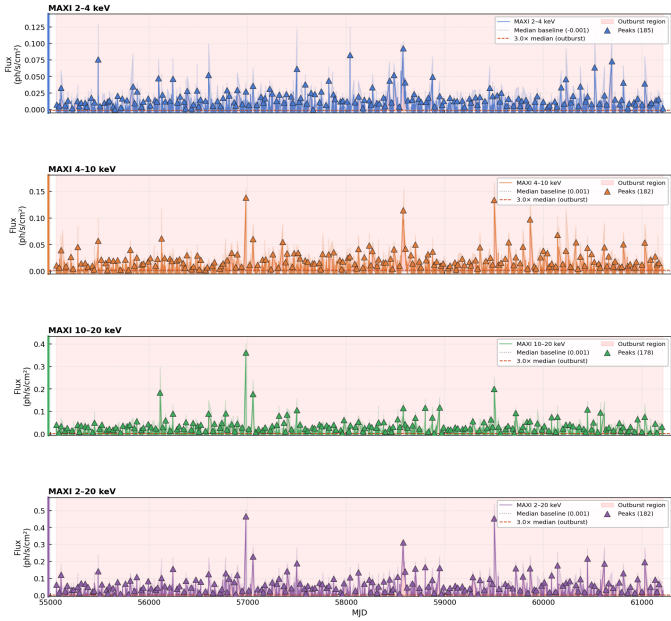
### Aql X-1

Aql X-1  
MAXI ID: J1911+005 · ZTF ID: ZTF18accedau



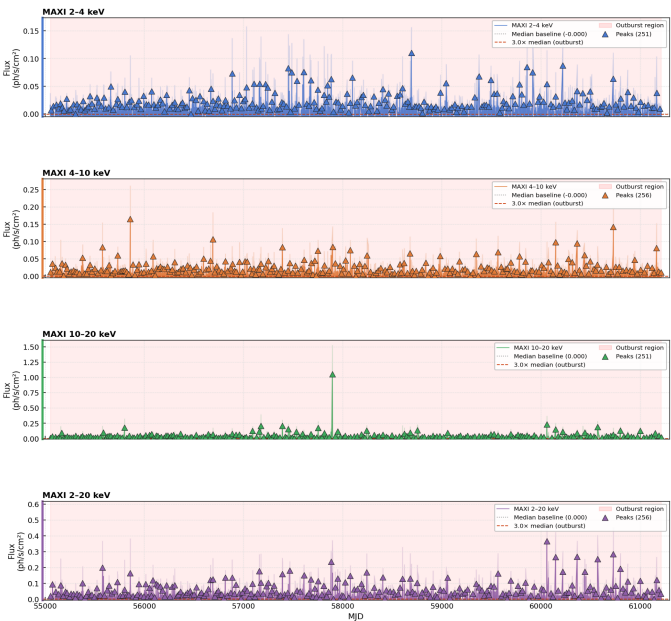
### GRO J2058+42

GRO J2058+42  
MAXI ID: J2058+42 · ZTF ID: ZTF18abc3th



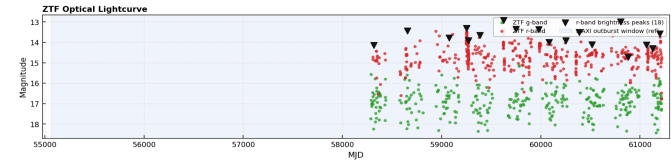
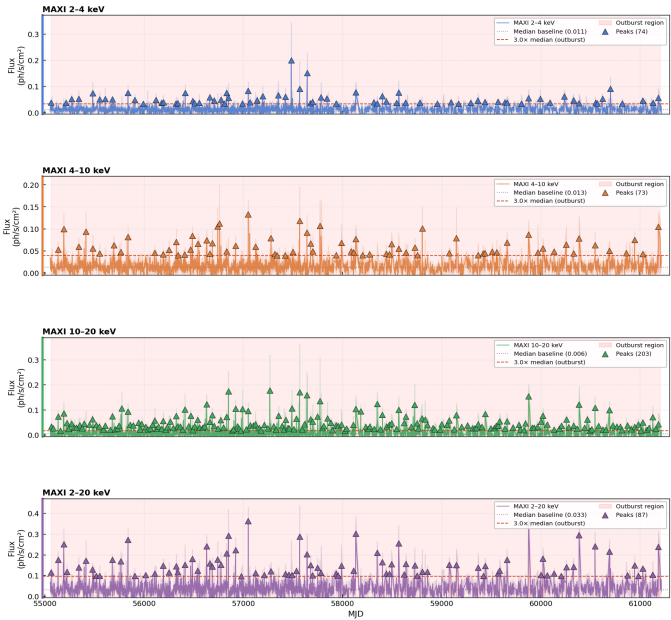
### 1A 0620-00

1A 0620-00  
MAXI ID: J0621-000 · ZTF ID: ZTF18acszzdo



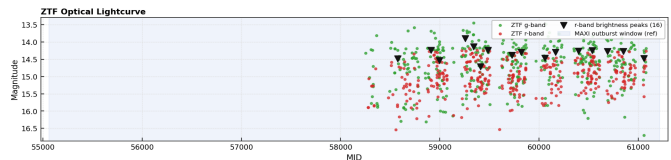
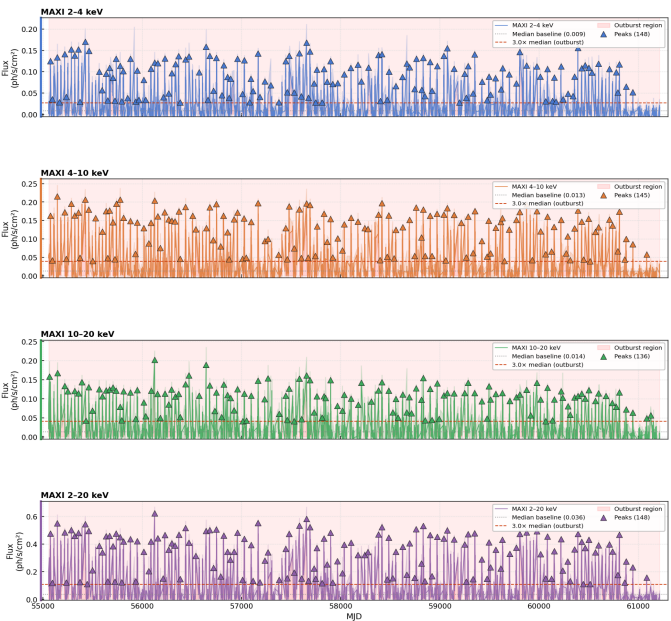
### SS 433

SS 433  
MAXI ID: J1911+049 · ZTF ID: ZTF18abc3thb



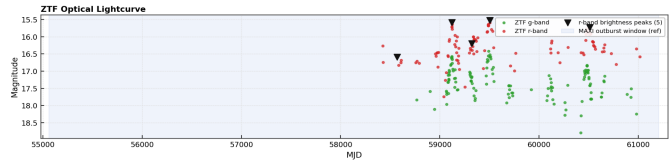
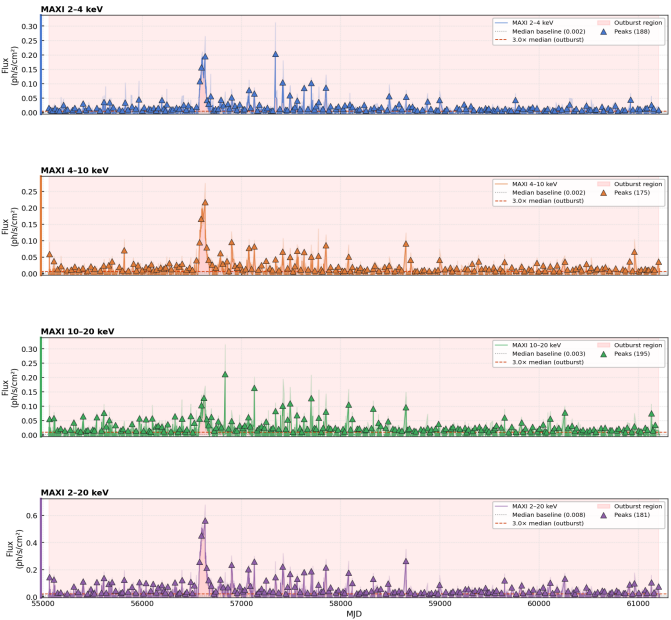
### Her X-1

Her X-1  
MAXI ID: J1657+537 · ZTF ID: ZTF18aatpzc



### KS 1947+300

KS 1947+300  
MAXI ID: J1949+302 · ZTF ID: ZTF18accozw



### 3.2 Classification of MAXI X-ray Binary Candidates Using SIMBAD, BlackCAT, and Liu et al. Catalogs

The MAXI GSC source list contains objects of heterogeneous astrophysical types. Before any timing or spectral analysis can be carried out, confirmed X-ray binaries (XRBs) must be isolated and their compact objects — black holes (BHs) or neutron stars (NSs) — identified. This section describes the two-stage pipeline that performs this task: a SIMBAD pre-filter that retains only XRB-type objects, followed by a positional and name-based cross-match against two curated reference catalogs to assign compact-object classes.

#### Databases

**MAXI GSC.** The Monitor of All-sky X-ray Image (MAXI) Gas Slit Camera (GSC) source catalogue provides all-sky X-ray source positions, fluxes, and light curves on day-to-month timescales.

**SIMBAD.** The SIMBAD astronomical database at CDS Strasbourg is queried to obtain standardised object-type labels for each MAXI position, enabling the rejection of non-XRB X-ray emitters such as active galactic nuclei and coronal stars.

**Liu et al. LMXB and HMXB Catalogs.** Two VizieR-hosted reference catalogs — the low-mass XRB (LMXB) catalog (J/A+A/469/807, 187 sources; Liu et al. 2007) and the high-mass XRB (HMXB) catalog (J/A+A/455/1165, 114 sources; Liu et al. 2006) — supply persistent/transient flags and, where available, compact-object identifications based on pulsation periods and spectroscopic mass functions.

**BlackCAT.** The Black-Hole Transients Catalog (Corral-Santana et al. 2016), maintained to  $\sim 2024$ , lists dynamically confirmed and candidate BH X-ray transients together with their sky positions and a curated set of cross-identification aliases ( $\sim 55$  entries).

#### Algorithm

The pipeline takes as input `step1_maxi_xrb_sources.csv`, which contains MAXI positions already filtered to XRB-type objects by a prior SIMBAD query. It appends four columns and writes the result to `step1b_maxi_xrb_classified.csv`:

- `compact_object` — BH, NS, or unknown
- `persistent_flag` — P (persistent), T (transient), or unknown
- `blackcat_name` — matched BlackCAT canonical name, if any
- `liu_name` — matched Liu et al. source name, if any

Classification follows a strict priority order applied to each source in sequence:

1. **BlackCAT match  $\Rightarrow$  BH.** The source is matched against BlackCAT by exact normalised name, then by substring, and finally by sky position within a 5 arcmin radius. A match immediately sets `compact_object` = BH.

2. **Liu et al. match  $\Rightarrow$  NS or P/T flag.** The same 5 arcmin position search is applied to the LMXB catalog first and then to the HMXB catalog. A non-empty pulsar-period column in the matched row sets `compact_object` = NS; the persistent/transient column sets `persistent_flag`. For HMXB NS pulsars, the persistent flag defaults to P when absent.
3. **Known-persistent override.** A hardcoded list of 28 sources with established persistent emission (e.g. Sco X-1, Cyg X-1, Crab) forces `persistent_flag` = P regardless of what Liu reports.
4. **BH heuristic.** For BH sources not matched by Liu, a simple rule assigns `persistent_flag` = P if the BlackCAT canonical name belongs to the set of five known persistent BHs (Cyg X-1, LMC X-1, LMC X-3, GRS 1758–258, 1E 1740.7–2942), and T otherwise, reflecting the near-universal transient behaviour of BH X-ray binaries.

The 5 arcmin cross-match radius is a practical compromise: it is large enough to tolerate the relatively coarse MAXI positional uncertainties ( $\lesssim 2'$  for bright sources) while remaining small enough to suppress spurious associations in the crowded Galactic-centre region.

## Name Normalisation

Source names in MAXI, SIMBAD, and the reference catalogs use inconsistent spacing and punctuation. All names are normalised to uppercase and stripped of spaces, hyphens, plus signs, and decimal points before comparison:

```
def _norm(s: str) -> str:
    return (str(s).upper()
            .replace(' ', '')
            .replace('-', '')
            .replace('+', '')
            .replace('.', ''))
```

Substring matching is then performed with a minimum key length of six characters to reduce false positives:

```
for bc_key, bc_canon in _bc_name_map.items():
    if len(bc_key) > 6:
        if bc_key in key or (len(key) > 6 and key in bc_key):
            return bc_canon
```

## Position Cross-match

Positional matching uses `astropy.coordinates.SkyCoord` and its `separation` method. For the BlackCAT match:

```

src = SkyCoord(ra=float(ra_deg)*u.deg, dec=float(dec_deg)*u.deg)
seps = src.separation(_bc_sc)          # _bc_sc: SkyCoord of all BlackCAT entries
idx = int(np.argmin(seps))
if seps[idx] < SEARCH_RADIUS:          # SEARCH_RADIUS = 5 arcmin
    return _bc_canon_list[idx]

```

An identical procedure is applied to the Liu LMXB and HMXB tables after loading them from VizieR with `astroquery`:

```

res = VizieR(columns=['**']).get_catalogs("J/A+A/469/807")
lmbx_table = res[0]

```

Because VizieR coordinate-column names are not standardised across catalogs, a helper function tries a prioritised list of candidate column names (`_RAJ2000`, `RAJ2000`, `RA`, etc.) and falls back to parsing sexagesimal strings when decimal degrees are absent.

### Classification Decision Logic

The full per-source decision block reflects the priority ordering described above:

```

# Step 1: BlackCAT -> BH
bc_match = match_blackcat(maxi_name, simbad_name, ra, dec)
if bc_match:
    compact_object = 'BH'
    blackcat_out    = bc_match

# Step 2: Liu et al. -> persistent_flag + compact_object
liu_pflag, liu_co, liu_name_val, liu_cat = query_liu(ra, dec)
if compact_object == 'unknown' and liu_co in ('BH', 'NS'):
    compact_object = liu_co
if liu_pflag in ('P', 'T'):
    persistent_flag = liu_pflag

# Step 3: Known-persistent override
for kp in _norm_persistent_set:
    if kp in _norm(maxi_name) or kp in _norm(simbad_name):
        persistent_flag = 'P'
        break

# Step 4: BH heuristic
if persistent_flag == 'unknown' and compact_object == 'BH':
    if bc_match in _PERSISTENT_BH_CANONS:
        persistent_flag = 'P'
    else:
        persistent_flag = 'T'

```

## Output and Downstream Use

The script writes a csv, which extends the SIMBAD-filtered source table with the four classification columns described above. A summary printed at runtime reports the counts in each class; representative numbers from a full run are given in Table 4 .

Table 4: Classification summary from the csv

| Category               | Sub-class      | Count |
|------------------------|----------------|-------|
| <b>compact_object</b>  | BH             | 30    |
|                        | NS             | 29    |
|                        | unknown        | 127   |
| <b>persistent_flag</b> | P (persistent) | 37    |
|                        | T (transient)  | 44    |
|                        | unknown        | 105   |
| Cross-match hits       | BlackCAT       | 30    |
|                        | Liu et al.     | 61    |

This classified table is the primary input to the next pipeline stage, which applies outburst-detection criteria to separate persistent emitters from transient sources and to flag sources suitable for Histogram analysis.

The bottom Table contains the persistent sources used in the Histogram study.

Table 5: Confirmed transient X-ray binaries used for Plotting Histograms.

| S.No. | MAXI Name    | SIMBAD Type | SIMBAD RA          | SIMBAD DEC         | CO | Liu Name         | BlackCAT Name |
|-------|--------------|-------------|--------------------|--------------------|----|------------------|---------------|
| 1     | 1A 0535+262  | HXB         | 84.72739538        | 26.31578804        | NS | 1A 0535+262      |               |
| 2     | 1A 0620-00   | HXB         | 95.68559327        | -0.345636239       | BH | 1A 0620-00       | A 0620-00     |
| 3     | 1A 1744-361  | LXB         | 267.0547833333333  | -36.13250556       | NS | 1A 1744-361      |               |
| 4     | 2S 1417-624  | HXB         | 215.30065010917747 | -62.69891371       | NS | 2S 1417-624      |               |
| 5     | 4U 0115+63   | HXB         | 19.633210398259997 | 63.74252017        | NS | 4U 0115+634      |               |
| 6     | 4U 0728-25   | HXB         | 112.2232451        | -26.10802698       | NS | 3A 0726-260      |               |
| 7     | 4U 1036-56   | HXB         | 159.3971438        | -56.79885267       | NS | RX J1037.5-5647  |               |
| 8     | 4U 1543-475  | HXB         | 236.78448694697997 | -47.6695235        | BH | 4U 1543-47       | 4U 1543-47    |
| 9     | 4U 1608-52   | LXB         | 243.179            | -52.4238           | NS | 4U 1608-52       |               |
| 10    | 4U 1630-472  | LXB         | 248.50670833333336 | -47.393            | BH | 4U 1630-47       | 4U 1630-47    |
| 11    | 4U 1711-34   | LXB         | 258.58241666666663 | -34.04647222       | NS | 2S 1711-339      |               |
| 12    | 4U 1901+03   | HXB         | 285.914136         | 3.2043772417300005 | NS | 4U 1901+03       |               |
| 13    | 4U 1907+09   | HXB         | 287.4085476        | 9.829797261        | NS | 4U 1907+09       |               |
| 14    | 4U 1957+115  | LXB         | 299.8500309        | 11.70829438        | BH | 4U 1957+11       | 4U 1957+11    |
| 15    | 4U 2238+60   | HXB         | 339.8368515        | 61.27405763        | NS | SAX J2239.3+6116 |               |
| 16    | AT 2019wey   | LXB         | 68.847             | 55.37618055555556  | BH |                  | AT2019wey     |
| 17    | Aql X-1      | LXB         | 287.8169047        | 0.5849633979699999 | NS | 4U 1908+005      |               |
| 18    | Cen X-4      | LXB         | 224.5913949        | -31.66875429       | NS | 4U 1456-32       |               |
| 19    | Cep X-4      | HXB         | 324.87787626970993 | 56.98622832        | NS | 1H 2138+579      |               |
| 20    | EXO 1846-031 | LXB         | 282.32125          | -3.062222222       | BH |                  | EXO 1846-031  |
| 21    | EXO 2030+375 | HXB         | 308.0636458        | 37.63745639578999  | NS | EXO 2030+375     |               |
| 22    | GRO J1008-57 | HXB         | 152.4456843        | -58.29322329       | NS | GRO J1008-57     |               |
| 23    | GRO J1655-40 | HXB         | 253.5005933179246  | -39.84576714       | BH | GRO J1655-40     | GRO J1655-40  |
| 24    | GRO J2058+42 | HXB         | 314.6980715        | 41.77699458        | NS | GRO J2058+42     |               |
| 25    | GRS 1716-249 | LXB         | 259.9038341        | -25.01781156       | BH | GRO J1719-24     | GRS 1716-249  |
| 26    | GRS 1747-312 | LXB         | 267.68975          | -31.29236111       | NS | GRS 1747-312     |               |

Continued on next page

Table 5 – Continued from previous page

| S.No. | MAXI Name                     | SIMBAD Type | SIMBAD RA          | SIMBAD DEC        | CO | Liu Name          | BlackCAT Name      |
|-------|-------------------------------|-------------|--------------------|-------------------|----|-------------------|--------------------|
| 27    | GRS 1915+105                  | HXB         | 288.798149         | 10.945807         | BH | GRS 1915+105      | GRS 1915+105       |
| 28    | GS 0834-430 - GS 0836-429     | HXB         | 128.9810823        | -43.18560925      | NS | GS 0834-430       |                    |
| 29    | GS 1124-683                   | HXB         | 171.6108173        | -68.67579938      | BH | GS 1124-684       | GS 1124-683        |
| 30    | GS 1354-64                    | HXB         | 209.5405113906579  | -64.73479385      | BH | GS 1354-64        | GS 1354-64         |
| 31    | GS 1826-238                   | LXB         | 277.3698999999999  | -23.7967          | NS | GS 1826-238       |                    |
| 32    | GS 1843+00                    | HXB         | 281.4034819        | 0.863177164       | NS | GS 1843+009       |                    |
| 33    | GS 1843-02                    | HXB         | 282.0737500000001  | -2.420277778      | NS | 2S 1845-024       |                    |
| 34    | GS 2000+251                   | HXB         | 300.70634183629625 | 25.23628657401223 | BH | GS 2000+25        | GS 2000+25         |
| 35    | GX 301-2                      | HXB         | 186.6565022        | -62.77035029      | NS | 4U 1223-624       |                    |
| 36    | GX 304-1                      | HXB         | 195.3212371        | -61.60184376      | NS | 4U 1258-61        |                    |
| 37    | GX 339-4                      | LXB         | 255.7057818        | -48.78974665      | BH | 4U 1659-487       | GX 339-4           |
| 38    | H 1145-619                    | HXB         | 177.0000894        | -62.20691733      | NS | 2S 1145-619       |                    |
| 39    | H 1658-298                    | LXB         | 255.52725000000004 | -29.94558333      | NS | MXB 1659-298      |                    |
| 40    | H 1730-333                    | LXB         | 263.34999999999997 | -33.3886          | NS | MXB 1730-335      |                    |
| 41    | H 1743-322                    | LXB         | 266.5649849        | -32.23357222      | BH | H1743-322         | H 1743-322         |
| 42    | HETE J1900.1-2455             | LXB         | 285.0360416666667  | -24.92047222      | NS | HETE J1900.1-2455 |                    |
| 43    | IGR J00291+5934 with V709 Cas | LXB         | 7.26275            | 59.57194444444445 | NS | IGR J00291+5934   |                    |
| 44    | IGR J11435-6109               | HXB         | 176.0012282        | -61.12680088      | NS | IGR J11435-6109   |                    |
| 45    | IGR J18410-0535               | HXB         | 280.251814         | -5.59624297       | NS | AX J1841.0-0536   |                    |
| 46    | KS 1947+300                   | HXB         | 297.3978505        | 30.20882666       | NS | KS 1947+300       |                    |
| 47    | MAXI J0556-332                | LXB         | 89.193             | -33.1745          | BH |                   | MAXI J0556-332     |
| 48    | MAXI J1659-152                | LXB         | 254.75700000000003 | -15.25798056      | BH |                   | MAXI J1659-152     |
| 49    | MAXI J1820+070                | LXB         | 275.09142659019994 | 7.185356905       | BH |                   | MAXI J1820+070     |
| 50    | MAXI J1910-057                | LXB         | 287.595            | -5.798861111      | BH |                   | Swift J1910.2-0546 |
| 51    | MXB 0656-072                  | HXB         | 104.5720326        | -7.209778968      | NS | XTE J0658-073     |                    |
| 52    | NGC 6440                      | LXB         | 267.2192708333333  | -20.36125         | NS | SAX J1748.9-2021  |                    |
| 53    | SAX J1712.6-3739              | LXB         | 258.1541           | -37.6451          | NS | SAX J1712.6-3739  |                    |

Continued on next page

Table 5 – Continued from previous page

| S.No. | MAXI Name          | SIMBAD Type | SIMBAD RA          | SIMBAD DEC         | CO | Liu Name           | BlackCAT Name      |
|-------|--------------------|-------------|--------------------|--------------------|----|--------------------|--------------------|
| 54    | SAX J1747.0-2853   | LXB         | 266.7608333333333  | -28.88302778       | NS | SAX J1747.0-2853   |                    |
| 55    | SAX J1808.4-3658   | LXB         | 272.11491666666666 | -36.97886111       | NS | SAX J1808.4-3658   |                    |
| 56    | SAX J1810.8-2609   | LXB         | 272.6852916666666  | -26.15033333       | NS | SAX J1810.8-2609   |                    |
| 57    | SAX J2103.5+4545   | HXB         | 315.8987920203899  | 45.75154688        | NS | SAX J2103.5+4545   |                    |
| 58    | Swift J1753.5-0127 | LXB         | 268.3678745        | -1.451737932       | BH | Swift J1753.5-0127 | Swift J1753.5-0127 |
| 59    | V 0332+53          | HXB         | 53.74963171        | 53.17313802        | NS | V 0332+53          |                    |
| 60    | V404 Cyg           | HXB         | 306.0159362655975  | 33.86721143190583  | BH | GS 2023+338        | V404 Cyg           |
| 61    | V4641 Sgr          | HXB         | 274.84014302468995 | -25.40718044       | BH | SAX J1819.3-2525   | SAX J1819.3-2525   |
| 62    | XTE J1118+480      | HXB         | 169.544971         | 48.03675409        | NS | XTE J1118+480      |                    |
| 63    | XTE J1543-568      | HXB         | 236.008            | -56.712            | NS | XTE J1543-568      |                    |
| 64    | XTE J1550-564      | HXB         | 237.7444325        | -56.47646066       | BH | XTE J1550-564      | XTE J1550-564      |
| 65    | XTE J1650-500      | HXB         | 252.5040833333333  | -49.96211111       | BH | XTE J1650-500      | XTE J1650-500      |
| 66    | XTE J1709-267      | LXB         | 257.377            | -26.6571           | NS | XTE J1709-267      |                    |
| 67    | XTE J1710-281      | LXB         | 257.5533           | -28.131            | NS | XTE J1710-281      |                    |
| 68    | XTE J1739-285      | LXB         | 264.9723           | -28.4952           | NS | XTE J1739-285      |                    |
| 69    | XTE J1752-223      | LXB         | 268.062875         | -22.34232222       | BH |                    | XTE J1752-223      |
| 70    | XTE J1807-294      | LXB         | 271.7491666666667  | -29.40833333       | NS | XTE J1807-294      |                    |
| 71    | XTE J1829-098      | HXB         | 277.43338          | -9.8564            | NS | XTE J1829-098      |                    |
| 72    | XTE J1855-026      | HXB         | 283.87669122807995 | -2.604649348       | NS | XTE J1855-026      |                    |
| 73    | XTE J1859+083      | HXB         | 284.7568125        | 8.245672222222222  | NS | XTE J1859+083      |                    |
| 74    | XTE J1859+226      | HXB         | 284.6728803        | 22.658270739999992 | BH | XTE J1859+226      | XTE J1859+226      |
| 75    | XTE J1908+094      | HXB         | 287.22116666666665 | 9.384677777777778  | BH | XTE J1908+094      | XTE J1908+094      |
| 76    | XTE J1946+274      | HXB         | 296.413987         | 27.36541712        | NS | XTE J1946+274      |                    |
| 77    | XTE J2012+381      | LXB         | 303.1573303        | 38.18354748        | BH | XTE J2012+381      | XTE J2012+381      |

## 4 XRB Outburst Detection

X-ray binary outbursts are transient episodes of dramatically enhanced accretion onto the compact object, producing X-ray luminosities that can rise several orders of magnitude above quiescence on timescales of days to weeks. Detecting and delineating these events in archival monitoring data requires distinguishing genuine flux excursions from instrumental noise and slow source variability. The pipeline described here operates on daily-binned light curves from the Monitor of All-sky X-ray Image (MAXI) instrument aboard the International Space Station, which covers the 2–20 keV band with a typical daily-bin sensitivity of  $\sim 10$  mCrab.

### 4.1 Manual Outburst Detection

The manual inspector provides an interactive workflow for careful examination of individual sources. The analyst fetches the light curve, visually identifies candidate outbursts against an  $N\sigma$  threshold, fits each candidate with an empirical model, and records the results.

#### Data acquisition

Given a MAXI source identifier (e.g. J1911+005), the one-day-binned all-band light curve is fetched directly from the MAXI public archive:

```
MAXI_ID = "J1911+005"
url = f"https://maxi.riken.jp/star_data/{MAXI_ID}/" \
      f"{MAXI_ID}_g_lc_1day_all.dat"
resp = requests.get(url, headers=headers, timeout=15)

lines = [l for l in resp.text.strip().split("\n")
          if l.strip() and not l.startswith("!")]
parsed = [(float(p[0]), float(p[1]))
           for l in lines for p in [l.split()] if len(p) >= 2]

mjd = np.array([p[0] for p in parsed])
flux = np.array([p[1] for p in parsed])
```

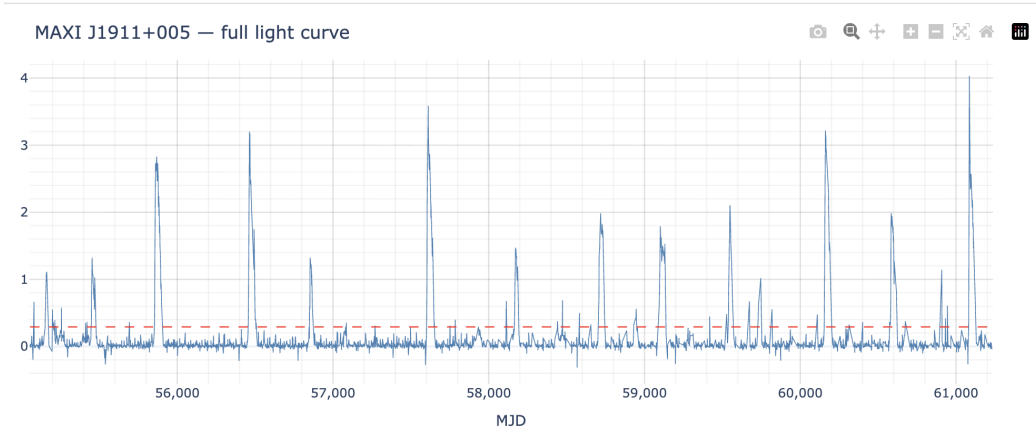
The result is a time series of Modified Julian Date (MJD) and flux in units of photons  $\text{s}^{-1} \text{cm}^{-2}$ .

#### Visual identification and threshold

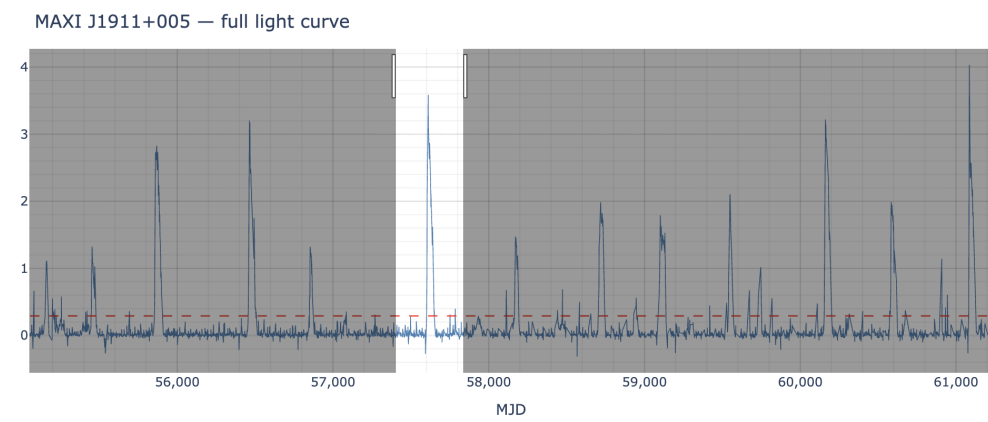
The full light curve is plotted with an  $N\sigma$  threshold overlaid to guide identification of candidate outbursts:

```
baseline = np.nanmedian(flux)
sigma = np.nanstd(flux[flux < np.nanpercentile(flux, 90)])
threshold = baseline + N_SIGMA * sigma # default N_SIGMA = 5
```

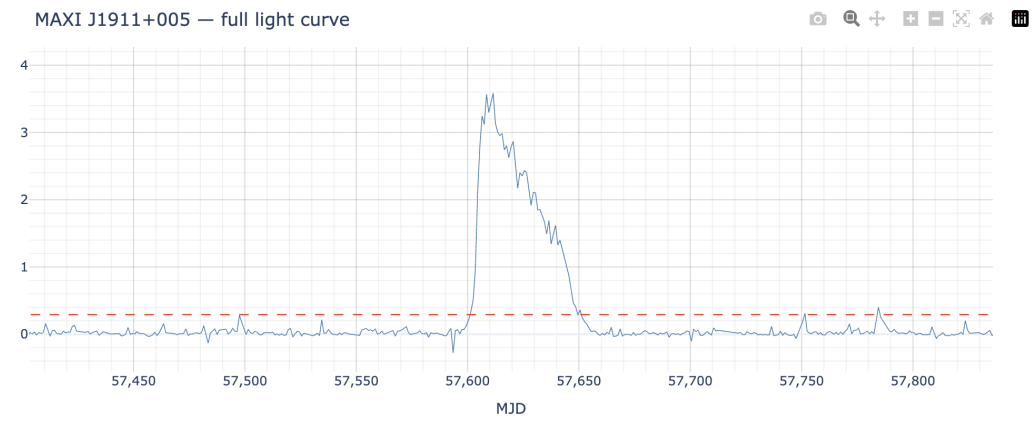
The quiescent standard deviation is estimated from the lower 90th percentile of the flux distribution to avoid contamination by outburst flux. The analyst zooms into suspected outburst windows and selects MJD bounds  $X1$ ,  $X2$  for fitting.



(a)



(b)



(c)

Figure 2: Three-panel illustration of the outburst-selection workflow for an example source **MAXI J1911+005**: the full MAXI light curve (a), the same curve with the dynamically-set 5 detection threshold and selected analysis window highlighted (b), and a zoomed view of a single outburst window (c) used as input to the FRED fitting pipeline..

## 4.2 Automated Outburst Detection

The automated pipeline processes the full XRB catalogue in a single run. Transient sources classified as LXB or HXB are read from `step1b_maxi_xrb_classified_full_tsv.csv`; persistently bright sources (flagged `persistent_flag == 'P'`) are skipped entirely, as their flux never returns to a true quiescent baseline. The remaining sources are routed to class-specific detection algorithms described below.

### Tier 0: viability filter

Before any outburst search, each source is tested against a global flux floor. If the peak MAXI flux across the entire light curve does not exceed  $0.04 \text{ photons s}^{-1} \text{ cm}^{-2}$ , the source is flagged as undetectable by MAXI in daily-binned data and excluded. This prevents spurious detections in catalogue sources that are too faint to be individually resolved by MAXI.

### Source routing

Sources passing Tier 0 are routed to one of four detection configurations based on their SIMBAD classification and compact-object type:

1. **BH LXB** — black-hole soft X-ray transients; long, bright, well-separated outbursts.
2. **NS LXB** — neutron-star transients; shorter and faster-recurrence outbursts with a valid spectral veto.
3. **Unknown LXB** — unclassified transients; treated conservatively.
4. **HXB** — high-mass X-ray binaries; persistent or semi-persistent emitters requiring a local rolling baseline.

The key numerical parameters for each configuration are listed in Table 6.

Table 6: Detection configuration parameters for each source class.

| Parameter                                            | BH LXB                    | NS LXB | Unk LXB | HXB                  |
|------------------------------------------------------|---------------------------|--------|---------|----------------------|
| Baseline method                                      | 15th pct (iterative clip) |        |         | 300-d rolling median |
| $\sigma$ multiplier                                  | 5.0                       | 5.0    | 5.5     | 5.0                  |
| Min contiguous days                                  | 10                        | 10     | 5       | 10                   |
| Min duration (d)                                     | 35                        | 25     | 30      | 25                   |
| Max duration (d)                                     | —                         | —      | —       | 200                  |
| Gap tolerance (d)                                    | 5                         | 3      | 3       | 3                    |
| Min peak flux ( $\text{ph s}^{-1} \text{ cm}^{-2}$ ) | 0.05                      | 0.08   | 0.05    | 0.10                 |
| Min separation (d)                                   | 120                       | 80     | 80      | 40                   |
| Merge gap (d)                                        | 80                        | 40     | 50      | 15                   |
| Max wing (d)                                         | 200                       | 120    | 150     | 80                   |
| Walk gap tolerance (d)                               | 5                         | 7      | 10      | 5                    |
| HR veto                                              | No                        | Yes    | No      | No                   |

## Iterative baseline estimation

For all LXB classes a robust quiescent baseline is required. A naive global percentile is biased upward in sources with many or long outbursts, because outburst flux contaminates the lower tail of the flux distribution. The pipeline instead uses an iterative sigma-clipping scheme:

```
def iterative_baseline(flux, percentile=15, sigma_clip=3.0, max_iter=10):
    mask = np.ones(len(flux), dtype=bool)
    for _ in range(max_iter):
        med = np.nanmedian(flux[mask])
        mad = np.nanmedian(np.abs(flux[mask] - med))
        sig = max(1.4826 * mad, 1e-6)
        mask = np.abs(flux - med) < sigma_clip * sig
        # stop when mask converges
    quiet_flux = flux[mask]
    baseline = np.nanpercentile(quiet_flux, percentile)
    mad_f = np.nanmedian(np.abs(quiet_flux - baseline))
    sigma_r = max(1.4826 * mad_f, 1e-6)
    return baseline, sigma_r
```

At each iteration, flux points more than  $3\sigma$  from the running median are masked out. The process repeats until the mask converges, after which the 15th percentile of the surviving (quiet) flux gives the baseline, and the robust scatter  $\sigma_r = 1.4826 \times \text{MAD}$  gives the noise width. The factor 1.4826 converts the MAD to an equivalent Gaussian  $\sigma$ .

## Peak finding and boundary delineation

Once the baseline and  $\sigma_r$  are established, the detection threshold is:

$$F_{\text{thr}} = F_{\text{base}} + N_{\sigma} \cdot \sigma_r \quad (4)$$

with  $N_{\sigma} = 5$  for LXBs. To identify peaks robustly, the raw light curve is first smoothed with a 7-day Savitzky–Golay filter (polynomial order 2) solely for the purpose of peak detection; the raw flux is used for everything else. Peaks are then located with `scipy.signal.find_peaks`:

```
smooth = savgol_filter(flux, window_length=7, polyorder=2)

peaks, props = find_peaks(
    smooth,
    height      = threshold,          # peak must clear 5-sigma
    prominence  = N_sigma * sigma_r, # must stand out from surroundings
    distance    = min_dist_pts,      # BH: 120 d apart; NS: 80 d
    width       = min_width_pts,     # must be >= MIN_CONTIGUOUS_DAYS
)
```

The **prominence** constraint is physically important: it requires each peak to rise by at least  $5\sigma_r$  above the highest flux in its immediate neighbourhood, preventing the algorithm from splitting a single outburst with a secondary maximum into two spurious events. The **distance** parameter encodes known recurrence timescales: at least 120 days for BH LXBs (where outbursts are separated by years of quiescence) and 80 days for NS LXBs (which can recur on shorter cycles).

Each detected peak is then assigned an outburst window by a boundary-walk algorithm. Starting from the interpolated half-width boundaries returned by `find_peaks`, the left boundary walks leftward and the right boundary walks rightward, extending as long as the flux remains above a lower threshold of  $F_{\text{base}} + 1\sigma_r$ . A gap tolerance of up to 5 days (BH) or 3 days (NS) is allowed to bridge short data dropouts. The walk is capped at `MAX_WING_DAYS` (200 d for BH, 120 d for NS) to prevent the boundary from absorbing a neighbouring outburst during a slow BH decay.

A candidate outburst is retained only if it satisfies:

1. Duration  $\geq$  `MIN_DURATION_DAYS` (35 d for BH LXB; 25 d for NS LXB).
2. Peak flux  $\geq$  `MIN_PEAK_FLUX` ( $0.05 \text{ ph s}^{-1} \text{ cm}^{-2}$  for BH;  $0.08 \text{ ph s}^{-1} \text{ cm}^{-2}$  for NS).

Two candidates whose windows overlap or are separated by less than `MERGE_GAP_DAYS` are merged into a single outburst record, with the brighter peak adopted as the event peak.

### Hardness-ratio veto (NS LXBs only)

NS LXBs are subject to an additional spectral veto. Genuine accretion outbursts in these systems are accompanied by a softening of the X-ray spectrum as the accretion disc fills in and the source enters the soft state. The hardness ratio is defined as:

$$\text{HR} = \frac{F_{4-10 \text{ keV}}}{F_{2-4 \text{ keV}}} \quad (5)$$

and is smoothed with a 20-day rolling median to suppress single-measurement noise. For a candidate to be confirmed, the HR measured near the flux peak (within the region where the flux exceeds 50% of the peak value) must satisfy:

$$\text{HR}_{\text{peak}} \leq 0.75 \times \text{HR}_{\text{quiet}} \quad (6)$$

where  $\text{HR}_{\text{quiet}}$  is the median HR during quiescent intervals. Candidates that do not soften sufficiently are rejected as hard-state flares or noise excursions. The veto is disabled when fewer than 50 quiescent days are available (insufficient baseline) or when the quiescent HR itself exceeds 0.55 (source is intrinsically hard even in quiescence, making the veto unreliable). The veto is not applied to BH LXBs because black-hole soft X-ray transients are spectrally soft throughout most of their outburst and the HR contrast with quiescence is unreliable as a discriminator.

## HXB detection

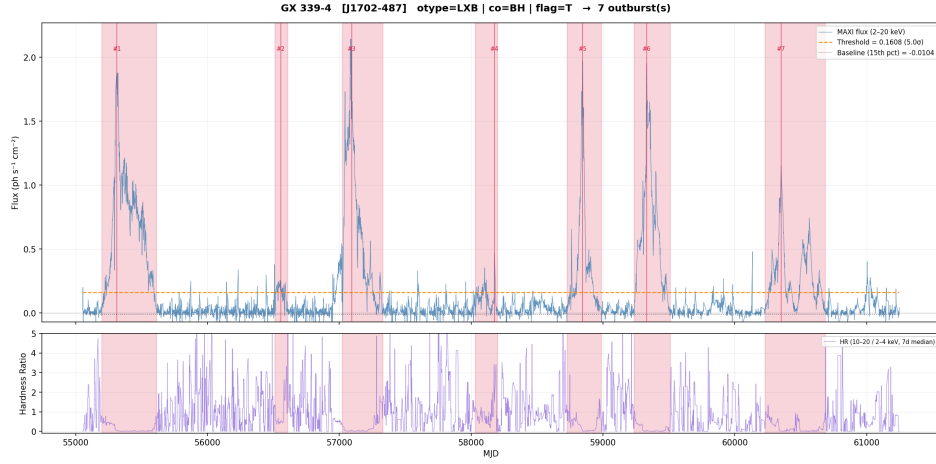
HXBs are persistent or semi-persistent emitters. Their flux never returns to a true quiescent floor, making a global percentile baseline meaningless. Instead, a 300-day rolling median of the 80th-percentile- clipped flux is used as a time-varying local baseline:

```
flux_clipped = np.clip(flux, None, np.nanpercentile(flux, 80))

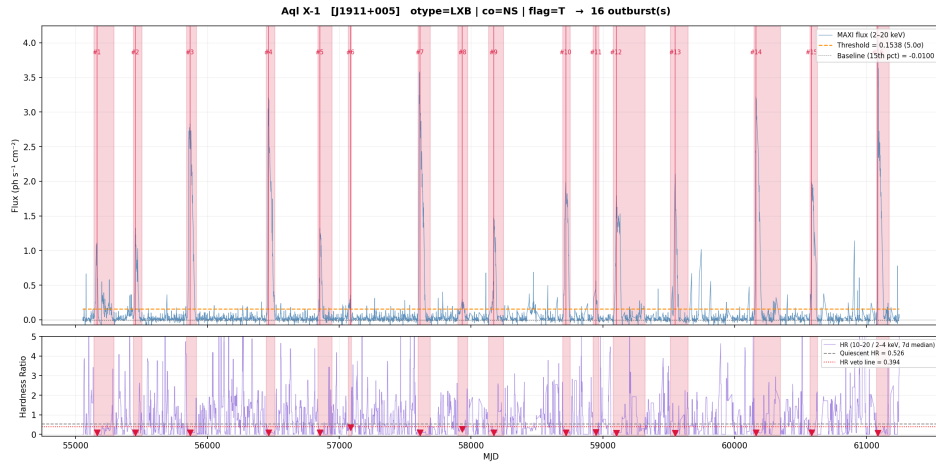
roll_med = pd.Series(flux_clipped).rolling(300, center=True,min_periods=100).median()
roll_mad = (pd.Series(flux_clipped)-roll_med).abs()
            .rolling(300, center=True, min_periods=100)
            .median()
sigma_r   = 1.4826 * roll_mad
thresh    = roll_med + 5.0 * sigma_r
```

Clipping at the 80th percentile before computing the rolling median prevents bright Type-I X-ray bursts or outburst peaks from biasing the local baseline upward. The resulting threshold is a smoothly varying curve that tracks the persistent emission level, so that discrete outbursts — which exceed it by  $5\sigma_r$  locally — can be identified against an inhomogeneous background.

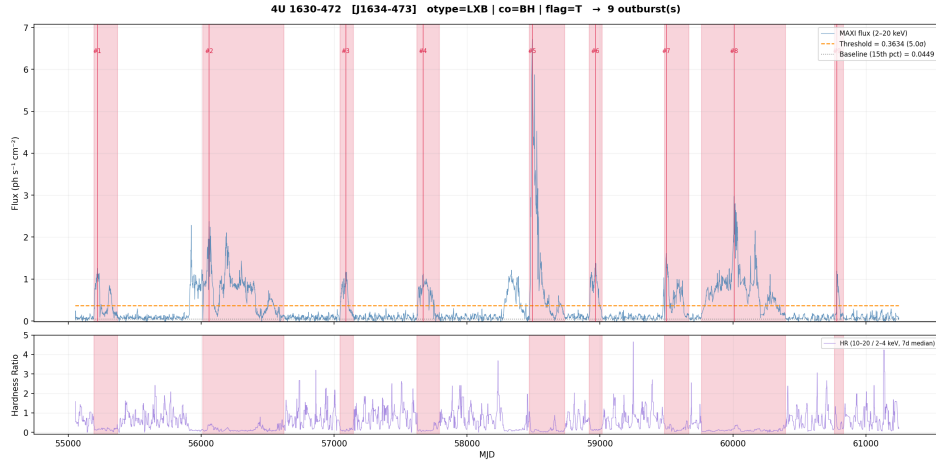
Peak finding and boundary walking then proceed identically to the LXB path, with parameters tuned for HXB variability: a minimum outburst separation of 40 days (matching typical Be/XRB orbital periods) and a duration ceiling of 200 days (events longer than this are classified as persistent emission episodes and excluded). No HR veto is applied, as HXBs are spectrally hard at all luminosity levels.



(a) GX 339-4 (BH)



(b) Aql X-1 (NS)



(c) 4U 1630-472 (BH)

Figure 3: Automated outburst detection applied to three well-studied X-ray binaries: (a) **GX 339-4 (BH)**, (b) **Aql X-1 (NS)**, and (c) **4U 1630-472 (BH)**, showing MAXI 2-20 keV light curves (*units converted to flux*) with  $5\sigma$  thresholds (orange dashed) and confirmed outburst windows (pink shading), alongside 7-day median hardness-ratio tracks (lower sub-panels), demonstrating the pipeline’s ability to recover outburst sequences across sources of differing compact-object type and recurrence behaviour.

## Pipeline outputs

Confirmed outbursts from all sources are written to a csv. Each row describes one outburst event; the output columns are listed in Table 7. Sources failing Tier 0 are recorded separately in another csv.

Table 7: Columns in `step2_outbursts.csv`.

| Column                       | Description                                                                     |
|------------------------------|---------------------------------------------------------------------------------|
| <code>maxi_name</code>       | MAXI source name from the input catalogue                                       |
| <code>t_start</code>         | MJD of outburst onset (left boundary walk end)                                  |
| <code>t_end</code>           | MJD of outburst end (right boundary walk end)                                   |
| <code>t_peak</code>          | MJD of peak flux within the outburst window                                     |
| <code>peak_flux</code>       | Peak flux [ $\text{ph s}^{-1} \text{cm}^{-2}$ ]                                 |
| <code>duration_days</code>   | Outburst duration = $t_{\text{end}} - t_{\text{start}}$ [d]                     |
| <code>baseline</code>        | Estimated quiescent baseline flux [ $\text{ph s}^{-1} \text{cm}^{-2}$ ]         |
| <code>threshold</code>       | Detection threshold flux [ $\text{ph s}^{-1} \text{cm}^{-2}$ ]                  |
| <code>quiescent_hr</code>    | Median hardness ratio in quiescent intervals (NS LXB only; NaN otherwise)       |
| <code>peak_hr</code>         | Near-peak hardness ratio (NS LXB only; NaN otherwise)                           |
| <code>hr_veto_used</code>    | Whether the HR spectral veto was active for this source                         |
| <code>source_type</code>     | Detector config used: HXB, LXB_BH, LXB_NS, or LXB_UNK                           |
| <code>source_id</code>       | Alphabetic source identifier (unique per source)                                |
| <code>outburst_id</code>     | Outburst identifier ( <code>source_id</code> + sequential integer, e.g. a1, a2) |
| <code>simbad_otype</code>    | SIMBAD object class (LXB or HXB)                                                |
| <code>compact_object</code>  | Compact object type (BH, NS, or unknown)                                        |
| <code>persistent_flag</code> | Source persistence flag (T = transient; unknown = unclassified)                 |

## 5 XRB Outburst FRED Fitting

X-ray binary outbursts recorded by MAXI are analysed using two complementary empirical profiles: a pure FRED and a FRED with a flat-top plateau (FRED+Tophat). Both are fitted to individual outburst peaks identified automatically from the 1-day all-band MAXI light curves. The pipeline proceeds in three stages for each source: light-curve retrieval, outburst detection, and least-squares profile fitting.

### 5.1 Pure FRED

#### Profile definition

A Fast Rise Exponential Decay (FRED) profile captures the asymmetric shape typical of X-ray transient outbursts: a rapid rise driven by accretion-disc instability followed by a slower exponential decay as the disc drains. The model has four free parameters,

$$F(t) = \begin{cases} A \exp\left(-\frac{t_{\text{peak}} - t}{\tau_{\text{rise}}}\right) & t < t_{\text{peak}}, \\ A \exp\left(-\frac{t - t_{\text{peak}}}{\tau_{\text{decay}}}\right) & t \geq t_{\text{peak}}, \end{cases} \quad (7)$$

where  $A$  is the peak flux,  $t_{\text{peak}}$  is the time of the peak (MJD),  $\tau_{\text{rise}}$  is the  $e$ -folding rise timescale, and  $\tau_{\text{decay}}$  is the  $e$ -folding decay timescale (both in days). A well-shaped outburst satisfies  $\tau_{\text{decay}} \gg \tau_{\text{rise}}$ .

#### Pipeline logic

**Stage 1 — Light-curve retrieval.** MAXI 1-day all-band light curves are fetched over HTTP from `maxi.riken.jp`. Comment lines (prefixed `!`) are stripped; the remaining rows are parsed as (MJD, flux) pairs. Sources returning fewer than 10 valid points are skipped.

**Stage 2 — Outburst detection.** A  $5\sigma$  threshold is computed from the quiescent baseline. The baseline is the median flux; the noise is the standard deviation of points below the 90th flux percentile (so bright outburst points do not inflate the noise estimate):

$$\theta = \text{median}(F) + 5 \sigma_{\text{noise}}. \quad (8)$$

Consecutive days above  $\theta$  are grouped into outburst episodes. Gaps of up to three days are bridged to handle MAXI data gaps. Groups spanning fewer than 25 days are rejected. Within each surviving group the dominant peak is identified with `scipy.signal.find_peaks`; secondary peaks within 10 days of a brighter one are suppressed.

**Stage 3 — FRED fitting.** A window of  $[-20, +50]$  days around each dominant peak is extracted. Initial guesses are derived from the data ( $A_0 = \max F_{\text{window}}$ ,  $t_{\text{peak},0}$  at that maximum,  $\tau_{\text{rise},0} = 3 \text{ d}$ ,  $\tau_{\text{decay},0} = 15 \text{ d}$ ). The model is fitted with `scipy.optimize.curve_fit` (Levenberg–Marquardt, `maxfev=10000`).

### Key code

```
def fred_model(t, A, t_peak, tau_rise, tau_decay):
    flux = np.where(
        t < t_peak,
        A * np.exp(-(t_peak - t) / tau_rise),
        A * np.exp(-(t - t_peak) / tau_decay)
    )
    return flux

def compute_threshold(flux, sigma_mult=5.0):
    baseline = np.nanmedian(flux)
    bulk_mask = flux < np.nanpercentile(flux, 90)
    noise_std = np.nanstd(flux[bulk_mask])
    return baseline + sigma_mult * noise_std, baseline, noise_std

popt, _ = curve_fit(
    fred_model, t_window, flux_window,
    p0=[A0, t_peak_guess, 3.0, 15.0],
    bounds=([0, t_min, 0.1, 0.1],
            [3*A0, t_max, days_before, 2*days_after]),
    maxfev=10000
)
```

#### 5.1.1 Outputs

Each successful fit produces (i) a diagnostic plot showing the MAXI data and the overlaid FRED model within the fit window, and (ii) a row in the summary table containing the source name, peak MJD,  $A$ ,  $\tau_{\text{rise}}$ ,  $\tau_{\text{decay}}$ , and the decay-to-rise ratio  $\tau_{\text{decay}}/\tau_{\text{rise}}$ .

### Plots for Pure FRED fit

A few of the Plots were given bellow for the source **4U 0115+634**.

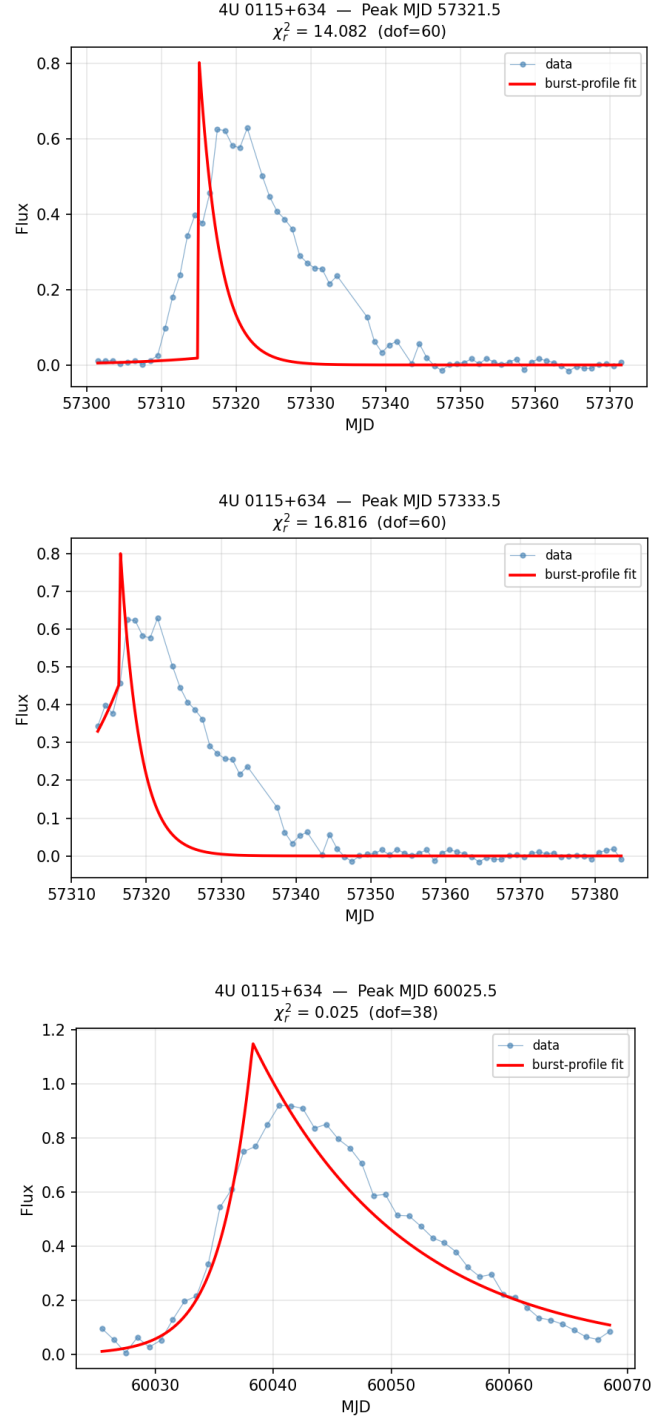


Figure 4: Three representative pure FRED (fast-rise exponential-decay) model fits to individual X-ray outbursts of the Be/X-ray binary **4U 0115+634**, illustrating a range of fit qualities  $\chi^2$  (reduced) ( 0.025 to 16.8) that expose the limitations of the symmetric FRED functional form for outbursts with complex or multi-peaked morphology.

## 5.2 FRED + Tophat

### Profile definition

Some outbursts exhibit a prolonged high-flux plateau between the rise and the decay. This is accommodated by inserting a flat segment of width  $w$  (days) between the two exponentials. The combined model has five free parameters,

$$F(t) = \begin{cases} A \exp\left(-\frac{t_s - t}{\tau_{\text{rise}}}\right) & t < t_s, \\ A & t_s \leq t \leq t_s + w, \\ A \exp\left(-\frac{t - (t_s + w)}{\tau_{\text{decay}}}\right) & t > t_s + w, \end{cases} \quad (9)$$

where  $t_s$  is the start of the plateau and  $w$  is its duration. Setting  $w = 0$  recovers the pure FRED (equation 28).

### Pipeline logic

Outburst detection is identical to the pure-FRED pipeline (Sections 8–28). The fitting stage differs in two respects.

First, the initial guess for  $t_s$  is placed at  $t_{\text{peak}} - w_0/2$  where  $w_0 = 10$  d, so the plateau is centred near the detected peak. Second, goodness of fit is assessed with a reduced chi-squared,

$$\chi_\nu^2 = \frac{1}{\nu} \sum_i \frac{(F_i - F_{\text{model},i})^2}{|F_i| + \epsilon}, \quad \nu = N - 5, \quad (10)$$

where  $\epsilon = 10^{-10}$  guards against zero-flux bins and  $\nu$  is the number of degrees of freedom ( $N$  data points, 5 free parameters). Comparing  $\chi_\nu^2$  between the pure FRED and FRED+Tophat fits for the same peak indicates whether the plateau component is statistically warranted.

### Key code

```
def fred_tophat_model(t, A, t_start, width, tau_rise, tau_decay):
    t_end = t_start + width
    flux = np.zeros_like(t)
    for i in range(len(t)):
        if t[i] < t_start: flux[i] = A * np.exp(-(t_start - t[i]) / tau_rise)
        elif t[i] <= t_end: flux[i] = A
        else: flux[i] = A * np.exp(-(t[i] - t_end) / tau_decay)
    return flux

def chi2_reduced(y_obs, y_model, n_params):
    chi2 = np.sum((y_obs - y_model)**2 / np.abs(y_obs + 1e-10))
    dof = max(len(y_obs) - n_params, 1)
    return chi2, dof, chi2 / dof

p0 = [A0, t_peak - w0/2, w0, 3.0, 15.0]
```

```

lower = [0, t_min, 0, 0.1, 0.1]
upper = [3*A0, t_max, t_max - t_min, 50, 200]

params, _ = curve_fit(
    fred_tophat_model, t_window, flux_window,
    p0=p0, bounds=(lower, upper), maxfev=10000
)

```

## Outputs

Each fitted peak yields a row containing  $A$ ,  $t_s$ ,  $w$ ,  $\tau_{\text{rise}}$ ,  $\tau_{\text{decay}}$ ,  $\chi^2$ , degrees of freedom, and  $\chi^2_\nu$ . Results are written to `fred_tophat_fit_summary.csv`.

## Plots for TopHat+FRED fit

A few of the TopHat + FRED Plots were given bellow for the source **4U 0115+634**.

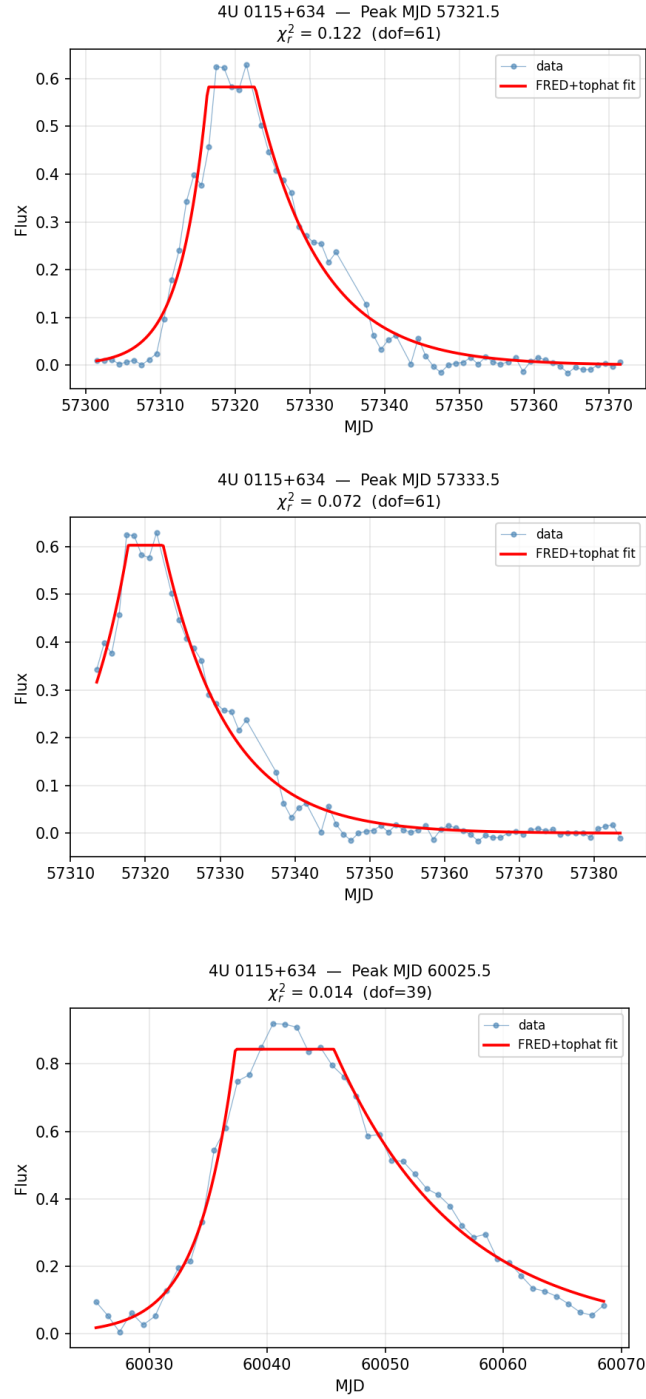


Figure 5: The same three outbursts of **4U 0115+634** now fitted with a TopHat+FRED composite model; a flat-topped "plateau" rise convolved with an exponential decay, yielding dramatically improved  $\chi^2$  (reduced) values (0.014-0.122 vs. 14-17 for the pure FRED), demonstrating that many XRB outbursts require a more flexible rise profile than a simple fast-rise function..

## 6 Histograms: XRB Outburst Duration Statistics

With a catalogue of detected outbursts in hand we characterise the temporal structure of each event through three duration metrics:  $T_{90}$ ,  $T_{\text{rise}}$ , and  $T_{\text{decay}}$ . All three are derived from the baseline-subtracted, cumulative flux profile of the MAXI light curve within the detected outburst window. Histograms of these quantities are constructed separately for black-hole (BH) and neutron-star (NS) systems to compare the outburst timescale distributions across compact-object classes.

### 6.1 $T_{90}$ : Total Outburst Duration

#### Definition and Mathematical Basis

$T_{90}$  is the interval over which an outburst accumulates the central 90% of its total net fluence. The metric is adapted from the same cumulative-flux formalism used for gamma-ray burst durations [e.g., 15].

For a detected outburst spanning  $[t_{\text{start}}, t_{\text{end}}]$ , the baseline-subtracted flux is

$$f_{\text{net}}(t_i) = \max(F(t_i) - F_{\text{base}}, 0), \quad (11)$$

where  $F(t_i)$  is the MAXI all-band flux at time  $t_i$  and  $F_{\text{base}}$  is the quiescent baseline estimated by iterative sigma-clipping. The cumulative fluence is

$$S(t_k) = \sum_{i=1}^k f_{\text{net}}(t_i). \quad (12)$$

The 5% and 95% fluence times,  $t_5$  and  $t_{95}$ , are defined by

$$S(t_5) = 0.05 S_{\text{total}}, \quad S(t_{95}) = 0.95 S_{\text{total}}, \quad (13)$$

and the outburst duration is

$$T_{90} = t_{95} - t_5. \quad (14)$$

#### Algorithm

1. Read the detected outburst table (`step2_outbursts.csv`) and the corresponding MAXI daily light curve for each source.
2. Mask the light curve to the outburst window  $[t_{\text{start}}, t_{\text{end}}]$  and subtract the baseline.
3. Clip negative net-flux values to zero and compute the cumulative sum.
4. Locate  $t_5$  and  $t_{95}$  by binary search on the cumulative array.
5. Record  $T_{90} = t_{95} - t_5$  in days.

The core Python implementation is:

```

def compute_t90(mjd, flux, baseline, t_start, t_end):
    mask = (mjd >= t_start) & (mjd <= t_end)
    if mask.sum() < 3:
        return np.nan, np.nan, np.nan
    ob_mjd = mjd[mask]
    ob_flux = flux[mask]
    net_flux = np.clip(ob_flux - baseline, 0, None)
    if net_flux.sum() == 0:
        return np.nan, np.nan, np.nan
    cumsum = np.cumsum(net_flux)
    total = cumsum[-1]
    t5_idx = np.searchsorted(cumsum, 0.05 * total)
    t95_idx = np.searchsorted(cumsum, 0.95 * total)
    t5_mjd = ob_mjd[t5_idx]
    t95_mjd = ob_mjd[t95_idx]
    t90 = t95_mjd - t5_mjd
    return t90, t5_mjd, t95_mjd

```

## Output and Usage

$T_{90}$  values are appended to the outburst catalogue as the column `t90_days` and saved to `step3_outbursts_t90.csv`. Histograms are plotted on a logarithmic time axis using 25 bins spanning 5–500 d (Figure 6). Dashed vertical lines mark the median  $T_{90}$  for each compact-object class. The BH and NS distributions are compared both with and without the unclassified (unknown) systems to assess whether classification uncertainty distorts the median.

## Plot

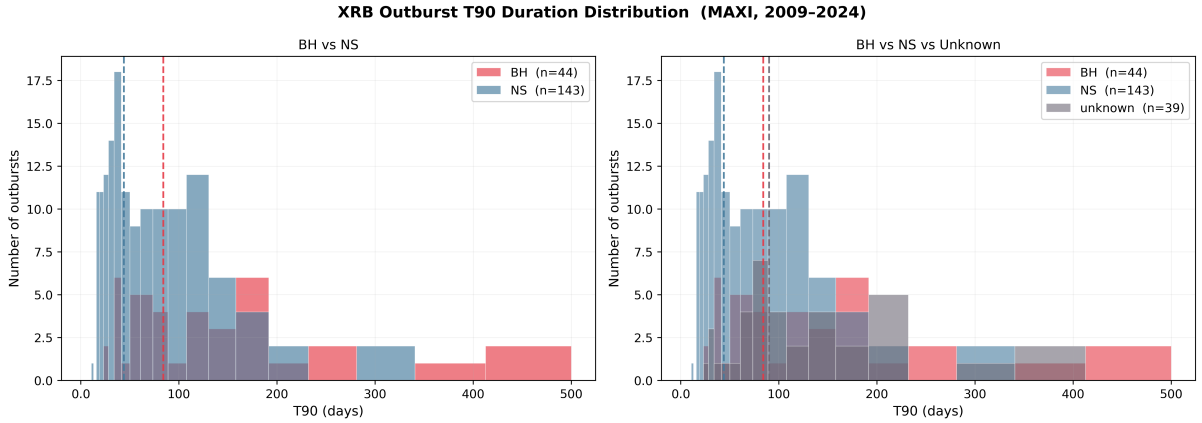


Figure 6:  $T_{90}$  outburst duration distributions for BH ( $n=44$ ) and NS ( $n=143$ ) systems detected by MAXI, plotted on a linear time axis; the clear rightward offset of the BH median relative to the NS median reflects their systematically longer outburst durations due to larger accretion discs that take more time to drain.

## 6.2 $T_{\text{rise}}$ and $T_{\text{decay}}$

### Definition and Mathematical Basis

$T_{\text{rise}}$  and  $T_{\text{decay}}$  partition  $T_{90}$  at the flux peak, measuring the rise and decline timescales independently. Within the  $[t_5, t_{95}]$  window, the peak is located at

$$t_{\text{peak}} = \arg \max_{t_5 \leq t_i \leq t_{95}} f_{\text{net}}(t_i). \quad (15)$$

The two sub-intervals are then

$$T_{\text{rise}} = t_{\text{peak}} - t_5, \quad T_{\text{decay}} = t_{95} - t_{\text{peak}}. \quad (16)$$

By construction,  $T_{\text{rise}} + T_{\text{decay}} = T_{90}$ . An asymmetry parameter,

$$\mathcal{A} = \frac{T_{\text{decay}} - T_{\text{rise}}}{T_{90}}, \quad (17)$$

ranges from  $-1$  (purely fast-rise slow-decay) to  $+1$  (purely slow-rise fast-decay), with  $\mathcal{A} = 0$  for a symmetric profile. XRB outbursts are typically fast-rise, exponential-decay (FRED), so  $\mathcal{A}$  is expected to be negative for most events.

### Algorithm

1. Reuse the cumulative flux arrays already computed for  $T_{90}$  (or recompute them from the same baseline-subtracted light curve).
2. Identify the peak index within  $[t_5, t_{95}]$ .
3. If the peak does not fall between  $t_5$  and  $t_{95}$  (which can occur for multi-peaked or pathological profiles), the event is flagged NaN and excluded from statistics.
4. Compute  $T_{\text{rise}}$ ,  $T_{\text{decay}}$ , and  $\mathcal{A}$ ; append to the catalogue.

The implementation is:

```
def compute_trise_tdecay(mjd, flux, baseline, t_start, t_end):
    mask = (mjd >= t_start) & (mjd <= t_end)
    if mask.sum() < 3:
        return np.nan, np.nan, np.nan
    ob_mjd = mjd[mask]
    ob_flux = flux[mask]
    net_flux = np.clip(ob_flux - baseline, 0, None)
    if net_flux.sum() == 0:
        return np.nan, np.nan, np.nan
    cumsum = np.cumsum(net_flux)
    total = cumsum[-1]
    t5_idx = np.searchsorted(cumsum, 0.05 * total)
    t95_idx = np.searchsorted(cumsum, 0.95 * total)
```

```

t5_mjd = ob_mjd[t5_idx]
t95_mjd = ob_mjd[t95_idx]
peak_idx = np.argmax(net_flux)
t_peak_mjd = ob_mjd[peak_idx]
if not (t5_mjd <= t_peak_mjd <= t95_mjd):
    return np.nan, np.nan, np.nan
return t_peak_mjd - t5_mjd, t95_mjd - t_peak_mjd, t_peak_mjd

```

## Output and Usage

$T_{\text{rise}}$  and  $T_{\text{decay}}$  are stored as `t_rise_days` and `t_decay_days` in `step3_outbursts_t90.csv` and later propagated to the master catalogue `step4_master_catalogue.csv`. A  $2 \times 2$  panel of histograms is produced (Figure 7): rise and decay distributions are shown side-by-side, each in two versions — BH vs. NS only, and BH vs. NS vs. unclassified. Logarithmic bins spanning 1–300 d are used. Median lines are overlaid per class. The ratio  $T_{\text{decay}}/T_{\text{rise}}$  and the asymmetry  $\mathcal{A}$  (Equation 17) are also recorded in the master catalogue for use in population studies.

## Plot

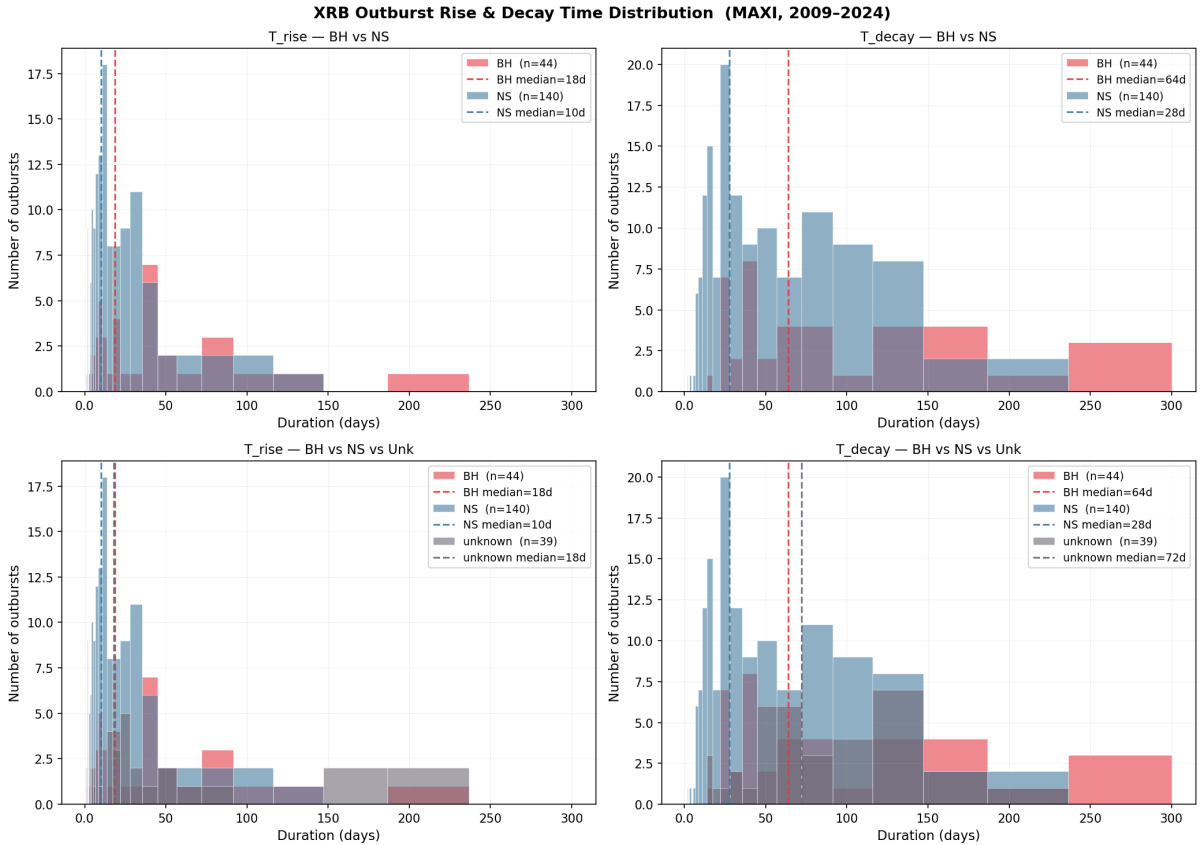


Figure 7:  $T_{\text{rise}}$  and  $T_{\text{decay}}$  distributions for BH (n=44) and NS (n=140) outbursts detected by MAXI, shown both as a BH-NS comparison (top) and with unclassified systems included (bottom); the clear rightward offset of the BH medians relative to NS in both rise (18d vs. 10d) and decay (64d vs. 28d) reflects their larger accretion discs that take longer to both ignite and drain, with the decay timescale being roughly 3-6 times longer than the rise timescale across all compact-object classes.

## 7 XRB Outburst Hardness

### 7.1 Hardness and the Hardness-Intensity Diagram

For an accreting X-ray binary, the *hardness ratio* is defined as the ratio of the source flux measured in a harder (higher-energy) band to the flux measured in a softer (lower-energy) band:

$$\text{HR} = \frac{F_{\text{hard}}}{F_{\text{soft}}} \quad (18)$$

A low HR indicates a spectrum dominated by soft, disk-like emission, while a high HR indicates a spectrum dominated by a hard, Comptonized component. The *intensity* is simply the total broadband flux (or count rate) of the source at that epoch.

A *Hardness-Intensity Diagram* (HID) is a scatter plot of intensity against hardness for a source, typically traced out over the course of an outburst [12]. Rather than looking at flux and hardness as two separate time series, the HID shows how the two evolve jointly, which is what actually carries information about the accretion state.

An analogous diagram can be constructed in the optical/IR using ZTF photometry. Since ZTF reports magnitudes rather than fluxes, a *color*, the analog of hardness, is used instead of a ratio:

$$C_{12} = F_1 - F_2 \quad (19)$$

where  $F_1$  and  $F_2$  are the fluxes (converted from AB magnitude) in two ZTF bands, and the corresponding optical intensity is taken as  $F_1 + F_2$ .

### 7.2 What the HID is used for

The HID is one of the standard diagnostic tools used to track the spectral evolution of an X-ray binary through an outburst. In practice it is used to:

- Identify spectral state transitions (hard  $\rightarrow$  soft  $\rightarrow$  hard) over the course of a single outburst, seen as distinct branches in the diagram rather than a random scatter of points.
- Distinguish accretion-disk-dominated emission (soft, low HR, high intensity) from corona/jet-dominated emission (hard, high HR).
- Compare the outburst morphology of black hole versus neutron star LXBs, and persistent versus transient HXBs, since the branches traced out in the HID differ systematically between source classes.
- Correlate the X-ray HID against the simultaneous optical HID from ZTF, to check whether optical brightening/reddening tracks the X-ray state (disk reprocessing, jet contribution, etc.).

### 7.3 Methodology

For a chosen outburst of a source, we fetch the corresponding MAXI (X-ray) and ZTF (optical) data restricted to that outburst's time window; the data fetching itself is covered in an earlier section. Given this data, the hardness and intensity are computed as follows.

### 7.3.1 X-ray hardness

For MAXI, hardness is computed using the 4–10 keV band as the hard band and the 2–4 keV band as the soft band, with intensity taken as the total 2–20 keV flux:

```
hr_x = flux_4_10 / (flux_soft + EPSILON)
intensity_x = flux_total
```

A small constant `EPSILON` is added to the denominator purely to avoid division by zero; it has no physical significance.

### 7.3.2 Optical color

ZTF detections in different bands are not simultaneous, so before a color can be computed, both bands are first collapsed to nightly averages and matched night-by-night:

```
def nightly_binned(df):
    d = df.copy()
    d['night'] = np.floor(d['mjd']).astype(int)
    return d.groupby('night').agg(mjd=('mjd', 'mean'),
                                  F_OIR=('F_OIR', 'mean')).reset_index()

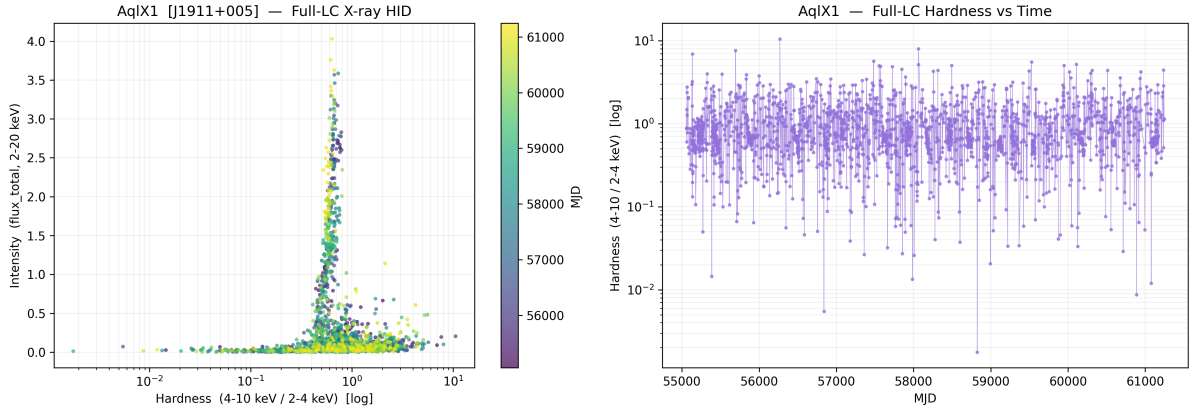
merged = nightly_binned(band1_df).merge(nightly_binned(band2_df), on='night')
color    = merged['F_OIR_band1'] - merged['F_OIR_band2']
intensity = merged['F_OIR_band1'] + merged['F_OIR_band2']
```

The band pair used is  $g-r$  when both bands have sufficient nightly coverage in the outburst window, falling back to  $r-i$  otherwise, since ZTF  $i$ -band coverage is frequently sparse.

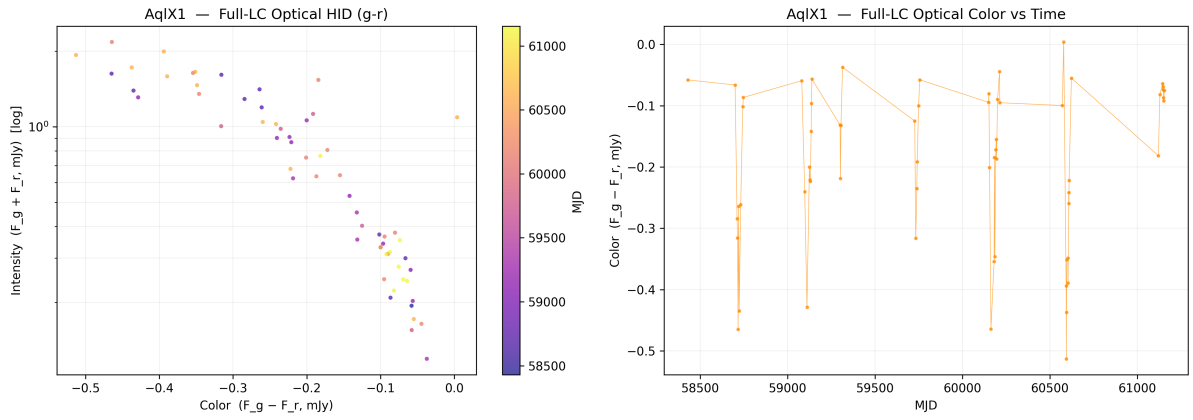
## 7.4 Plots

For each inspected outburst, the notebook produces up to four plots: an X-ray HID and an X-ray hardness-versus-time plot (always, since MAXI coverage is assumed available), and an optical HID and optical color-versus-time plot (only when a usable ZTF band pair exists for that outburst window; otherwise these are skipped). Example plots for the Aql X-1 outburst AqlX1\_2 are shown below.

## Global Plots



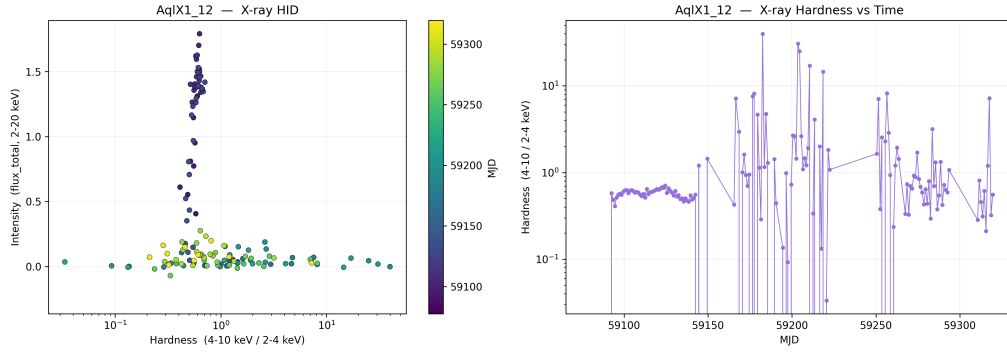
(a) MAXI (X-ray) global light curve.



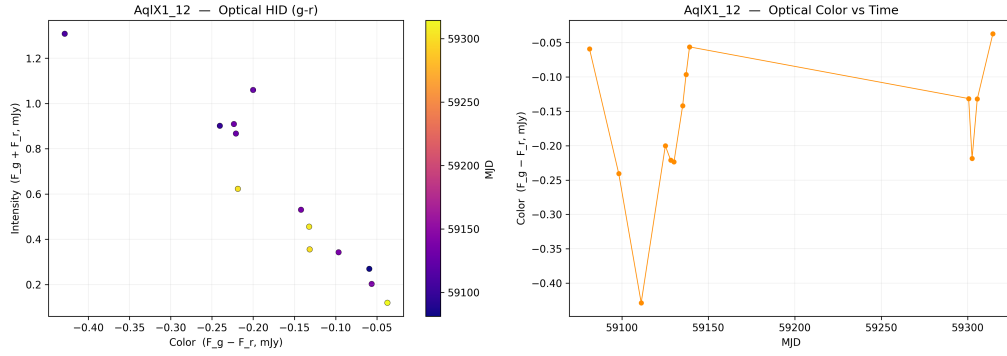
(b) ZTF (optical) global light curve.

Figure 8: Full-baseline hardness-intensity diagrams (HIDs) and time-series hardness tracks for Aql X-1 in both MAXI X-ray (top) and ZTF optical g-r colour (bottom), colour-coded by MJD.

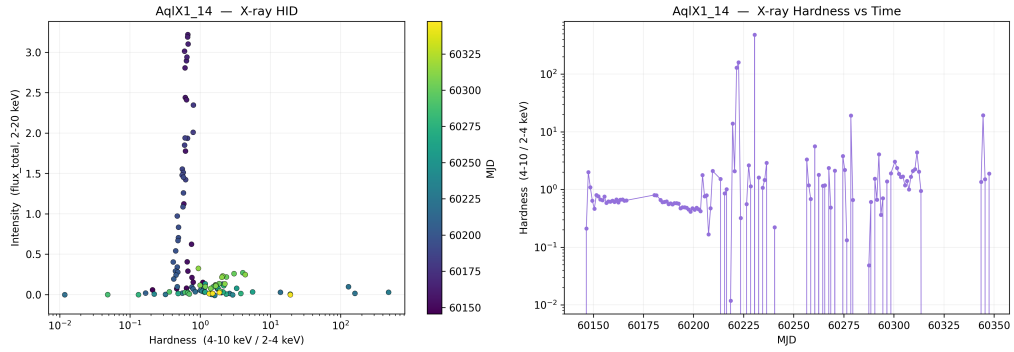
## Outburst Specific Plots



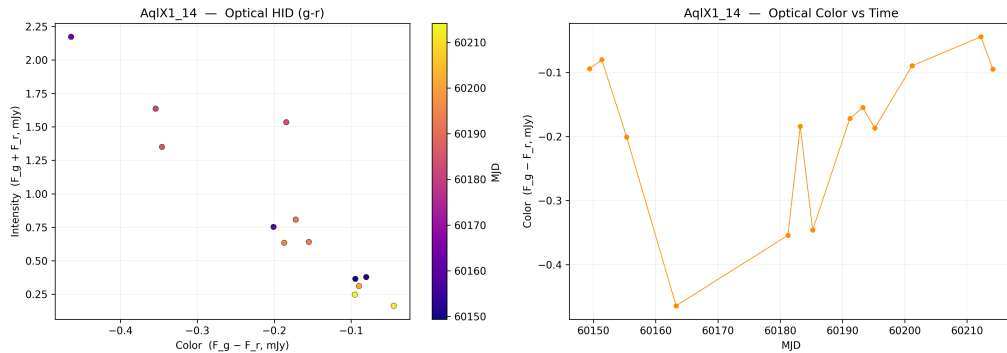
(a) Outburst AqlX1\_12 – MAXI (X-ray).



(b) Outburst AqlX1\_12 – ZTF (optical).



(c) Outburst AqlX1\_14 – MAXI (X-ray).



(d) Outburst AqlX1\_14 – ZTF (optical).

Figure 9: MAXI and ZTF views of the two commonly detected outbursts of Aql X-1: AqlX1\_12 (rows 1–2) and AqlX1\_14 (rows 3–4).

## 8 Modeling Optical Outburst from X-Ray data

X-ray binary outbursts produce broadband emission spanning from X-ray to optical and infrared wavelengths. A fraction of the hard X-ray flux irradiates the accretion disc and the companion star, is reprocessed, and re-emitted at longer wavelengths. This coupling imposes a power-law relationship between the optical/infrared (OIR) flux and the X-ray flux:

$$F_{\text{OIR}} = C_{\text{source}} \cdot F_X^\beta \quad (20)$$

where  $\beta$  encodes the physical mechanism responsible for the optical emission and  $C_{\text{source}}$  is a source-specific normalisation that absorbs distance, inclination, and disc geometry. The present work implements this framework numerically and applies it to outbursts detected in MAXI X-ray light curves, with simultaneous ZTF optical photometry. The two main analysis stages are described in the subsections below: one where  $\beta$  is held fixed at literature values and  $C_{\text{source}}$  is fitted, and one where both  $\beta$  and  $C_{\text{source}}$  are free parameters.

### 8.1 Physical and Observational Framework

#### The reprocessing relation and its context

Russell et al. [28] compiled quasi-simultaneous OIR and X-ray data from 33 X-ray binaries spanning 8 orders of magnitude in X-ray luminosity and showed that equation (20) holds as a global correlation in the hard spectral state. The slope  $\beta$  is not universal; its value encodes the dominant emission process.

For X-ray reprocessing in the disc, van Paradijs [32] showed that the optical luminosity of an irradiated disc scales as  $L_{\text{OPT}} \propto L_X^{0.5}$ , giving  $\beta = 0.5$  as the theoretical reprocessing prior. Russell et al. [28] confirmed this model fits the optical NSXB data well, where X-ray reprocessing is the established dominant mechanism. For the empirical global fits to their full sample, they report:

$$L_{\text{OIR}} = 10^{13.1 \pm 0.6} L_X^{0.61 \pm 0.02} \quad (\text{BH LMXBs, hard state}) \quad (21)$$

$$L_{\text{OIR}} = 10^{10.8 \pm 1.4} L_X^{0.63 \pm 0.04} \quad (\text{NS LMXBs, hard state}) \quad (22)$$

The two slopes are remarkably similar, but the BHXB normalisation is on average  $\sim 19.4$  times larger than the NSXB normalisation at a given  $L_X$ . For BHXBs, Russell et al. [28] find that both X-ray reprocessing and jet emission predict a slope close to 0.6, and both processes contribute in the hard state, with the jets producing approximately 90 per cent of the NIR emission at high luminosities. For NSXBs, X-ray reprocessing alone can describe the optical correlations.

The viscously heated disc predicts a much flatter dependence:  $\beta \approx 0.15$ – $0.25$  for BHXBs and  $\beta \approx 0.30$ – $0.50$  for NSXBs [28], and is ruled out as the dominant process for BHXBs in the hard state. The jet model for BHXBs, assuming a flat radio-to-OIR spectrum, predicts  $\beta \approx 0.7$  [28].

In the soft state the NIR emission is suppressed relative to the hard-state correlation, consistent with the jet switching off at state transition. Calibration in the present pipeline is therefore

restricted to hard-state epochs only (Section 8.1).

In this work, two fixed values of  $\beta$  are used as reference runs following Russell et al. [28]:  $\beta_A = 0.50$  (the theoretical X-ray reprocessing prior from van Paradijs 32) and  $\beta_B = 0.61$  (the empirical BH LMXB best-fit). A third value,  $\beta_C = 0.78$ , was used as an additional reference based on a colleague’s single-outburst empirical fit and is not a global literature anchor.

### X-ray light curve and Crab conversion

MAXI (Monitor of All-sky X-ray Image) provides daily-binned light curves in units of counts  $\text{s}^{-1} \text{cm}^{-2}$  across the 2–20 keV band. To convert to physical energy flux we use the Crab nebula as a standard candle. The adopted conversion factor is:

$$C_{\text{Crab}} = \frac{2.4 \times 10^{-8}}{3.3} \text{ erg cm}^{-2} \text{ s}^{-1} (\text{ct s}^{-1} \text{ cm}^{-2})^{-1} \quad (23)$$

so that the calibrated X-ray energy flux at each epoch is

$$F_X = f_{\text{MAXI}} \times C_{\text{Crab}} \quad [\text{erg cm}^{-2} \text{ s}^{-1}] \quad (24)$$

In code this appears as:

```
MAXI_CONV = 2.4e-8 / 3.3    # ct/s/cm^2 -> erg/cm^2/s

F_X_raw    = flux_w        * MAXI_CONV
F_X_err_raw = np.abs(flux_err_w) * MAXI_CONV
```

### Hardness ratio and spectral-state classification

The MAXI 1-day file carries nine columns: MJD, 2–20 keV flux, and separate sub-band fluxes at 2–4, 4–10, and 10–20 keV (each with its error). The hardness ratio is defined as

$$\text{HR} = \frac{I(4\text{--}10 \text{ keV})}{I(2\text{--}4 \text{ keV})} \quad (25)$$

and each epoch is classified using fixed thresholds:

| State        | Condition               | Use in pipeline       |
|--------------|-------------------------|-----------------------|
| Hard         | $\text{HR} \geq 0.6$    | Calibration + fitting |
| Intermediate | $0.3 < \text{HR} < 0.6$ | Excluded from fits    |
| Soft         | $\text{HR} \leq 0.3$    | Excluded from fits    |

Table 8: Spectral-state classification from the MAXI hardness ratio. Only hard-state epochs enter calibration because the OIR–X-ray reprocessing relation of equation (20) applies specifically to that regime [28].

The guard against a negative 2–4 keV background-subtracted flux (which would make HR diverge or flip sign) is explicit:

```
soft_ok = maxi_df["flux_2_4"] > 0
```

```

maxi_df["hr"] = np.where(soft_ok,
    maxi_df["flux_4_10"] / maxi_df["flux_2_4"],
    np.nan)

def hardness_ratio_state(hr, hr_hard=0.6, hr_soft=0.3):
    if hr >= hr_hard: return "hard"
    elif hr <= hr_soft: return "soft"
    return "intermediate"

```

### Optical flux from ZTF magnitudes

ZTF reports PSF-fit AB magnitudes in the  $g$ ,  $r$ , and  $i$  bands. These are converted to flux density in mJy using the AB system zero-point of 3631 Jy:

$$F_{\text{OIR}} = 3631 \times 10^3 \times 10^{-0.4m} \quad [\text{mJy}] \quad (26)$$

with propagated uncertainty

$$\sigma_{F_{\text{OIR}}} = 0.4 \ln 10 \cdot F_{\text{OIR}} \sigma_m \quad (27)$$

In code:

```

F_OIR      = 3631000.0 * 10 ** (-0.4 * mag)      # mJy
F_OIR_err  = 0.4 * np.log(10) * F_OIR * mag_err

```

All three bands are pooled for calibration and plotted separately for visual inspection.

### X-ray outburst profile: the FRED model

Each outburst identified manually in the MAXI light curve is fitted with a Fast-Rise Exponential-Decay (FRED) profile:

$$F(t) = \begin{cases} A \exp\left[2\sqrt{\frac{t_{\text{rise}}}{t_{\text{decay}}}}\right] \exp\left[-\frac{t_{\text{rise}}}{t - t_{\text{start}}} - \frac{t - t_{\text{start}}}{t_{\text{decay}}}\right] & t > t_{\text{start}} \\ 0 & t \leq t_{\text{start}} \end{cases} \quad (28)$$

The four fitted parameters are the onset time  $t_{\text{start}}$ , rise sharpness  $t_{\text{rise}}$  (in days; smaller  $\Rightarrow$  sharper), exponential decay scale  $t_{\text{decay}}$ , and true peak amplitude  $A$ . The peak time follows analytically:

$$t_{\text{peak}} = t_{\text{start}} + \sqrt{t_{\text{rise}} t_{\text{decay}}} \quad (29)$$

and  $F(t_{\text{peak}}) = A$  exactly by construction — the profile is smooth everywhere with no kink at the peak. All four fitted values are stored in a JSON database and are not re-fitted in later pipeline phases; downstream code simply evaluates equation (28) at arbitrary times.

```

def fred_model(t, A, t_start, t_rise, t_decay):

```

```

t = np.atleast_1d(np.asarray(t, dtype=float))
model = np.zeros_like(t)
active = t > t_start
dt = t[active] - t_start
norm = A * np.exp(2.0 * np.sqrt(t_rise / t_decay))
model[active] = norm * np.exp(-t_rise / dt - dt / t_decay)
return model

```

## JSON database structure

All results are accumulated in a single JSON file keyed by MAXI source ID. Each source carries a list of outburst entries; each entry holds the FRED fit parameters, spectral-state calibration constants, lag times, and (after Phase 5) the free- $\beta$  results. A representative entry looks as follows:

```

{
  "J1911+005": {
    "display_name": "Aql X-1",
    "compact_object": "NS",
    "ztf_id": "ZTF18accedau",
    "outbursts": [
      {
        "status": "fitted",
        "t_start": 61082.34, "t_rise": 1.87,
        "t_decay": 18.6, "amplitude": 0.034,
        "C_A": 15.231, "C_A_err": 0.042,
        "C_B": 15.118, "C_B_err": 0.039,
        "tau_lag": -2.1, "N_calib": 12,
        "RMS_A": 0.00314, "RMS_B": 0.00287
      }
    ]
  }
}

```

## 8.2 Fixed- $\beta$ Calibration of $C_{\text{source}}$

### Overview

With  $\beta$  fixed at literature values —  $\beta_A = 0.50$  (theoretical reprocessing prior; and  $\beta_B = 0.61$  (empirical BH LMXB best-fit; Russell et al. 28) — the intercept  $C_{\text{source}}$  is computed analytically for each outburst using only hard-state epochs. This is Phases 1–4 of the pipeline. Note that Russell et al. [28] work in luminosity space ( $L_{\text{OIR}} \propto L_X^\beta$ ); in this pipeline we work directly in flux space ( $F_{\text{OIR}} \propto F_X^\beta$ ), with  $C_{\text{source}}$  absorbing the distance and normalisation.

## Temporal lag

The reprocessed optical emission responds to X-ray driving on a timescale of order days (viscous or light-travel delay). The optical flux is therefore compared against the FRED model evaluated at a lag-corrected time:

$$F_X^{\text{eff}}(t) = F_{\text{FRED}}(t - \tau_{\text{lag}}) \quad (30)$$

The lag  $\tau_{\text{lag}}$  is defined as the difference in MJD between the ZTF brightness peak (brightest point across all bands) and the X-ray FRED peak:  $\tau_{\text{lag}} = t_{\text{peak,OIR}} - t_{\text{peak,X}}$ . A positive value means the optical peak follows the X-ray peak; a negative value means the optical leads.

## Calibration estimator

For each epoch  $i$  in the hard state and within  $\Delta t \leq 1$  day of a ZTF observation, define the per-point intercept:

$$C_i = \log_{10}(F_{\text{OIR},i}) - \beta \log_{10}(F_{X,i}) \quad (31)$$

The weighted mean with inverse-variance weights  $w_i = \sigma_{C_i}^{-2}$  gives the best-fit intercept:

$$C_{\text{source}} = \frac{\sum_i w_i C_i}{\sum_i w_i}, \quad \sigma_{C_{\text{source}}} = \sqrt{\frac{1}{\sum_i w_i}} \quad (32)$$

where  $\sigma_{C_i}$  is propagated from the ZTF flux error via

$$\sigma_{C_i} = \frac{\sigma_{F_{\text{OIR},i}}}{F_{\text{OIR},i} \ln 10} \quad (33)$$

This is computed twice — once for  $\beta_A$  (yielding  $C_A$ ,  $\sigma_{C_A}$ ) and once for  $\beta_B$  (yielding  $C_B$ ,  $\sigma_{C_B}$ ) — and both are stored in the JSON. The code implementing equation (32) is:

```
ln10 = np.log(10.0)

logFOIR = np.log10(cal_FOIR)
logFX    = np.log10(cal_FX)
sigC     = cal_sOIR / (cal_FOIR * ln10)    # d(log10 F)/dF
w        = 1.0 / sigC**2

C_i = logFOIR - BETA_A * logFX            # per-point intercepts
C_A = np.sum(w * C_i) / np.sum(w)
C_A_err = np.sqrt(1.0 / np.sum(w))
```

## Optical simulation and RMS comparison

Once  $C_A$  and  $C_B$  are known, the simulated OIR flux at any time is

$$F_{\text{sim}}(t) = 10^C \cdot [F_{\text{FRED}}(t - \tau_{\text{lag}})]^\beta \quad (34)$$

This is evaluated on a dense time grid and overplotted against the observed ZTF points. The match quality is quantified by the root-mean-square residual on the pooled ZTF bands:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_{\text{OIR},i} - F_{\text{sim}}(t_i))^2} \quad [\text{mJy}] \quad (35)$$

The two values  $\text{RMS}_A$  and  $\text{RMS}_B$  are stored in the JSON and used in Phase 4 to compare which  $\beta$  assumption fits each outburst better across the full source sample.

```
F_X_at_ztf = fred_smooth_erg(ztf_df_all["mjd"].values - tau_lag)
```

```
F_sim_A = 10**ob["C_A"] * F_X_at_ztf**BETA_A    # mJy
```

```
F_sim_B = 10**ob["C_B"] * F_X_at_ztf**BETA_B    # mJy
```

```
resid_A = ztf_df_all["F_OIR"].values - F_sim_A
```

```
RMS_A    = float(np.sqrt(np.mean(resid_A**2)))
```

### Dual-axis plot for fixed $\beta$

Given are the plots for the source Aql-X1. Each outburst is represented by a dual-axis plot of the MAXI, FRED, ZTF, and reprocessing light curves.

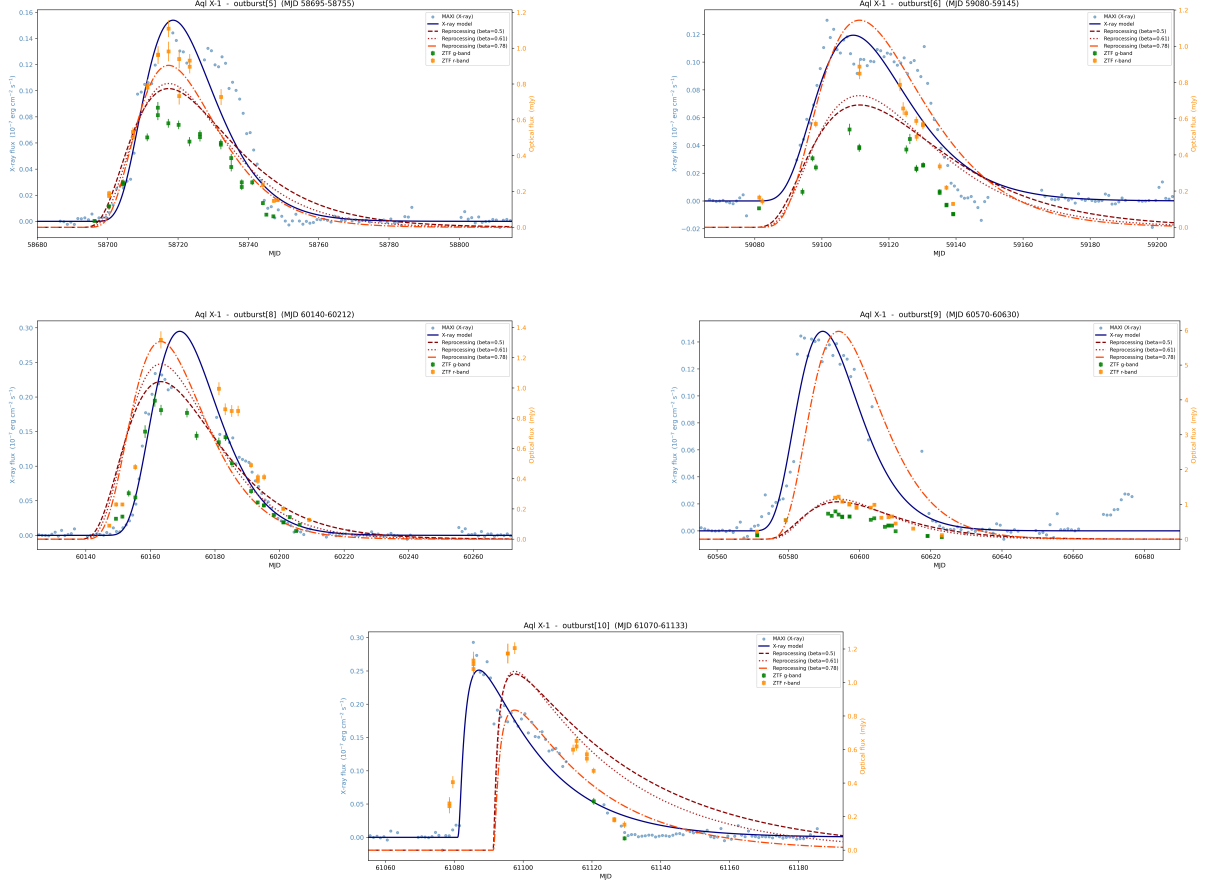


Figure 10: Dual-axis light curves for five Aql X-1 outbursts with the optical free-fit curve generated using a fixed X-ray-to-optical power-law index  $\beta$ , overlaying MAXI data, FRED model, ZTF observations, and the reprocessing-scaled optical prediction on a common time axis.

### 8.3 Free- $\beta$ Fitting

#### Motivation

The fixed- $\beta$  calibration of Section 8.2 assumes the reprocessing slope is known a priori. Phase 5 relaxes this assumption and solves for  $\beta$  and  $C_{\text{source}}$  simultaneously from the observed hard-state pairs. The result,  $\beta_{\text{free}}$ , can then be compared against the empirical anchors of Russell et al. [28] to assess whether a given outburst is consistent with standard disc reprocessing, jet-dominated emission, or some intermediate regime.

#### Weighted least squares in log-log space

Taking the logarithm of equation (20):

$$y_i = C + \beta x_i, \quad y_i \equiv \log_{10}(F_{\text{OIR},i}), \quad x_i \equiv \log_{10}(F_{X,i}) \quad (36)$$

This is a straight line in log-log space. With per-point weights  $w_i = \sigma_{y_i}^{-2}$ , where  $\sigma_{y_i} = \sigma_{F_{\text{OIR},i}} / (F_{\text{OIR},i} \ln 10)$ , the analytic weighted-least-squares solution is:

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i}, \quad \bar{y} = \frac{\sum w_i y_i}{\sum w_i} \quad (37)$$

$$\beta_{\text{free}} = \frac{\sum w_i (x_i - \bar{x})(y_i - \bar{y})}{\sum w_i (x_i - \bar{x})^2}, \quad C_{\text{free}} = \bar{y} - \beta_{\text{free}} \bar{x} \quad (38)$$

with formal uncertainties

$$\sigma_{\beta_{\text{free}}} = \sqrt{\frac{1}{\sum w_i (x_i - \bar{x})^2}}, \quad \sigma_{C_{\text{free}}} = \sqrt{\frac{\sum w_i x_i^2}{(\sum w_i) \sum w_i (x_i - \bar{x})^2}} \quad (39)$$

The system is solved analytically with no iterative solver. A `scipy.optimize.curve_fit` call on the same data serves as a numerical cross-check; both must agree to floating-point precision.

### Data selection and guards

The fitting dataset is rebuilt from scratch for each outburst: every ZTF observation is paired with the nearest hard-state MAXI epoch (within  $\Delta t \leq 1$  day), and the pair is discarded if either flux is non-positive before the log-transform (background-subtracted MAXI fluxes can dip to zero or below at low count rates).

```
hard_maxi = maxi_df[maxi_df["state"] == "hard"]

for _, zrow in ztf_df_all.iterrows():
    dt = np.abs(hard_maxi["mjd"] - zrow["mjd"])
    j = dt.idxmin()
    if dt.loc[j] > DT_MATCH: # DT_MATCH = 1.0 day
        continue
    cal_FOIR.append(zrow["F_OIR"])
    cal_FX.append(hard_maxi.loc[j, "flux_erg"])
    cal_sOIR.append(zrow["F_OIR_err"])

valid = (cal_FX > 0) & (cal_FOIR > 0) & (cal_sOIR > 0)
x = np.log10(cal_FX[valid])
y = np.log10(cal_FOIR[valid])
sigC = cal_sOIR[valid] / (cal_FOIR[valid] * np.log(10))
w = 1.0 / sigC**2
```

### Sample-size gate

A two-parameter fit from fewer than 5 points is unreliable; the pipeline skips it and retains the fixed- $\beta$  results. For  $5 \leq N_{\text{pairs}} < 8$  the fit proceeds with a printed caveat. A second guard catches the degenerate case where all surviving  $x_i$  are numerically identical (zero dynamic range in  $\log_{10} F_X$ ), which would cause a division by zero in the slope formula:

```
MIN_N_BETA = 5
```

```
if N_pairs < MIN_N_BETA:
```

```
    beta_free = None    # skip; fixed-beta C_A/C_B remain valid
```

```
elif Sxx <= 0:
```

```
    beta_free = None    # zero x-range; slope not identifiable
```

```
else:
```

```
    x_bar = np.sum(w * x) / np.sum(w)
```

```
    y_bar = np.sum(w * y) / np.sum(w)
```

```
    Sxx = np.sum(w * (x - x_bar)**2)
```

```
    Sxy = np.sum(w * (x - x_bar) * (y - y_bar))
```

```
    beta_free = Sxy / Sxx
```

```
    C_free = y_bar - beta_free * x_bar
```

```
    sigma_beta_free = np.sqrt(1.0 / Sxx)
```

```
    sigma_C_free = np.sqrt(np.sum(w * x**2) / (np.sum(w) * Sxx))
```

## Dual-Axis Plots for free $\beta$

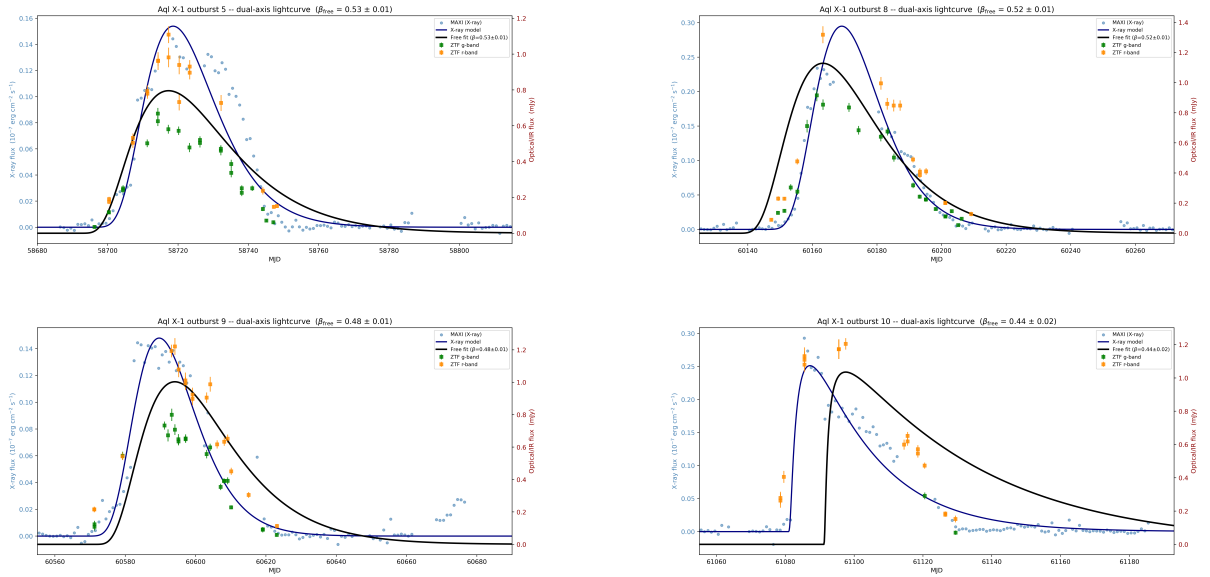


Figure 11: Dual-axis light curves for the four Aql X-1 outbursts (5,8,9,10) with  $\beta$  allowed to vary freely in the fit, yielding be,st-fit values of 0.53, 0.52, 0.48, and 0.44 respectively, with the black free-fit curve tracking the ZTF data more faithfully across rise, peak, and decay

## 9 Conclusion

This work presents a multiwavelength framework for X-ray binary outbursts, spanning source identification through optical–X-ray flux modelling. Starting from the Swift/BAT hard X-ray catalogue (932 sources), sequential cross-matching against MAXI, SIMBAD, and ZTF yielded 20 confirmed XRBs with simultaneous X-ray and optical coverage. A larger MAXI-only sample of 186 transients; 30 black-hole and 29 neutron-star systems, was classified using BlackCAT and the Liu et al. catalogues.

Outbursts were identified by comparing each source’s brightness against its own quiet-state baseline, flagging significant deviations, and using X-ray information to rule out false positives in neutron-star system. Detected outbursts were characterised by FRED and FRED+Tophat profile fits, recovering rise and decay timescales consistent with disc-instability driven accretion. Cumulative-fluence duration metrics ( $T_{90}$ ,  $T_{\text{rise}}$ ,  $T_{\text{decay}}$ ) reveal that BH systems have systematically longer outbursts (median  $T_{\text{decay}} \approx 64$  d) compared to NS systems (median  $T_{\text{decay}} \approx 28$  d), reflecting deeper quiescence and larger disc reservoirs in BH transients.

Hardness–intensity diagrams constructed from MAXI sub-band fluxes and ZTF optical colours confirm spectral softening during peak accretion, consistent with canonical hard-to-soft state transitions. Finally, the optical–X-ray flux correlation  $F_{\text{OIR}} = C_{\text{source}} F_X^\beta$  was calibrated per outburst in both fixed- $\beta$  ( $\beta_A = 0.50$ ,  $\beta_B = 0.61$ ) and free- $\beta$  regimes. Fitted slopes cluster near 0.5–0.6, consistent with X-ray disc reprocessing as the dominant optical emission mechanism during the hard state, with per-source scatter suggesting secondary jet contributions in several BH LMXBs. The pipeline is fully automated and reproducible, providing a foundation for population-level outburst studies as MAXI and ZTF baselines continue to expand.

Beyond its immediate application to the current MAXI–ZTF sample, this framework is directly extensible to the substantially larger cross-matched datasets that next-generation surveys will produce. As the time baselines of MAXI and ZTF lengthen, the number of detectable outbursts and archival cross-matches will grow correspondingly, allowing the statistical samples underlying the classification and duration analyses presented here to be refined with tighter constraints. More significantly, the automated cross-matching and outburst-characterisation steps developed in this work are well suited to the data volumes anticipated from the Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST), which will deliver optical light curves for billions of sources at a cadence and depth far exceeding ZTF, dramatically increasing the number of X-ray binary optical counterparts available for joint analysis. Similarly, the Nancy Grace Roman Space Telescope’s wide-field infrared surveys will extend this multiwavelength approach into a regime less affected by interstellar extinction, particularly valuable for XRBs situated in the heavily obscured regions of the Galactic plane and bulge where many such systems reside. Adapting the present pipeline to ingest Rubin/LSST and Roman data streams would allow automated, near-real-time identification and characterisation of X-ray binary outbursts across a much larger and more complete Galactic population, extending the population-level statistics on rise/decay timescales and optical–X-ray flux correlations presented in this work to a far greater number of sources.

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